



# *Digital Integrated Circuits*

## *A Design Perspective*

Jan M. Rabaey  
Anantha Chandrakasan  
Borivoje Nikolic

# Semiconductor Memories

*December 20, 2002*

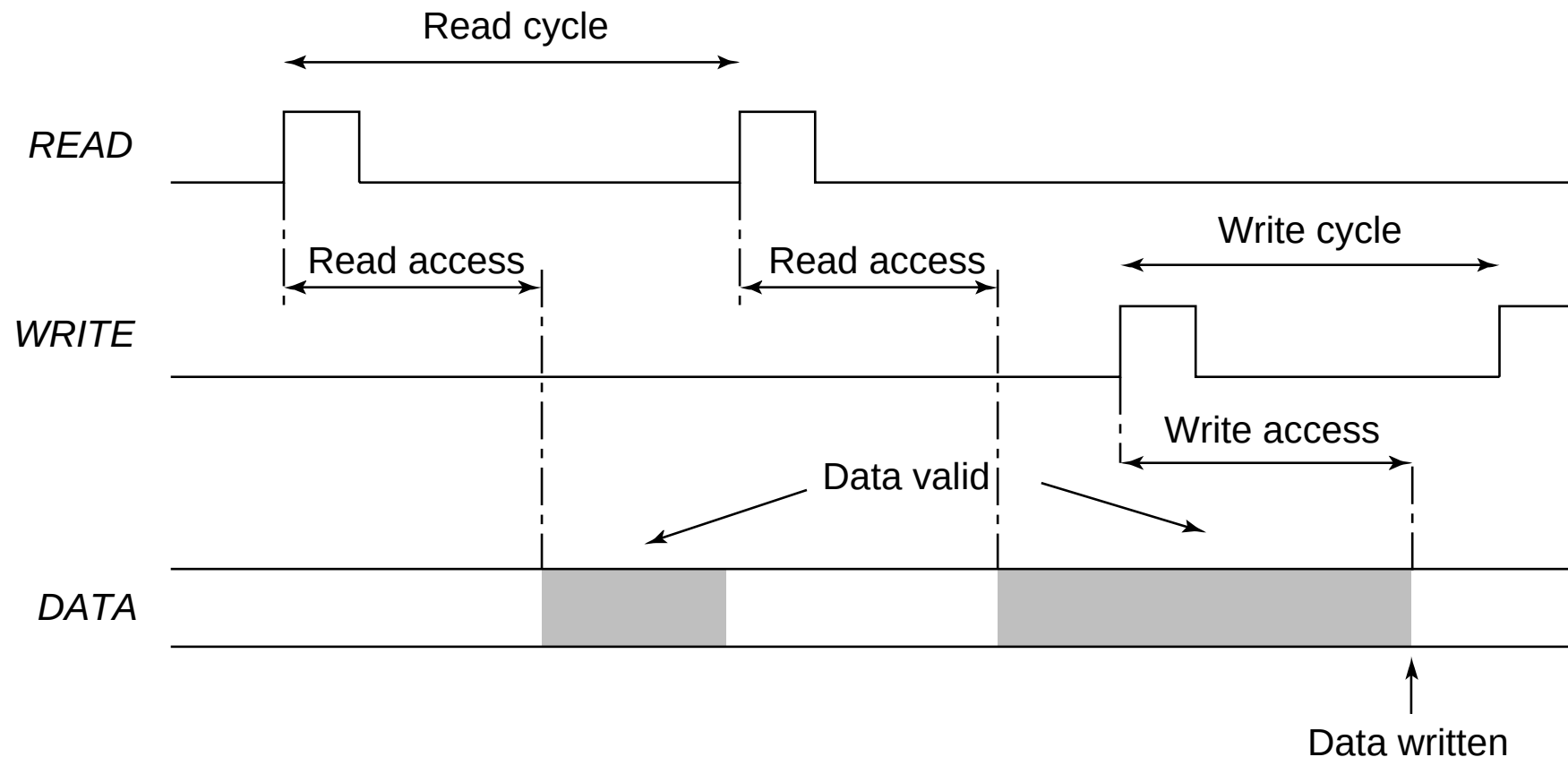
# ***Chapter Overview***

- ☐ **Memory Classification**
- ☐ **Memory Architectures**
- ☐ **The Memory Core**
- ☐ **Periphery**
- ☐ **Reliability**
- ☐ **Case Studies**

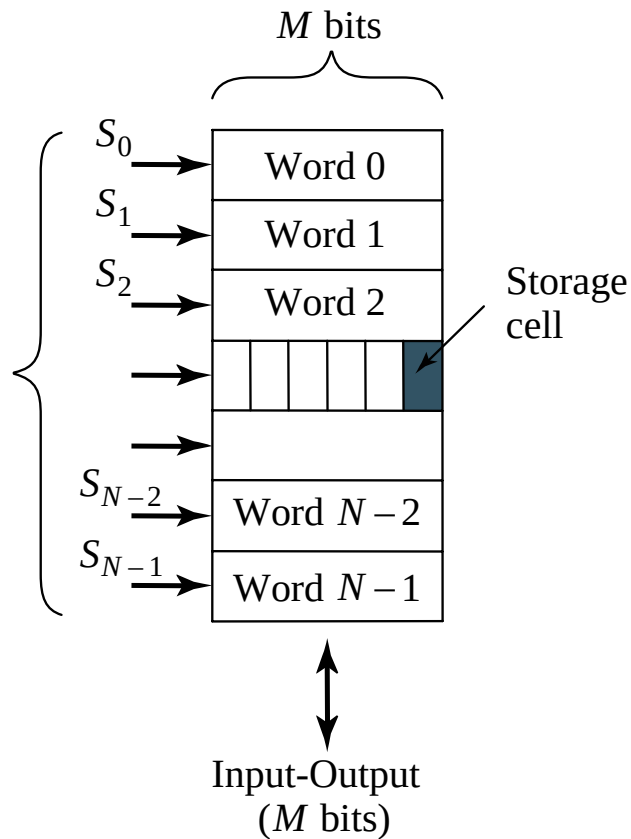
# Semiconductor Memory Classification

| Read-Write Memory |                                       | Non-Volatile Read-Write Memory        | Read-Only Memory                    |
|-------------------|---------------------------------------|---------------------------------------|-------------------------------------|
| Random Access     | Non-Random Access                     | EPROM<br>E <sup>2</sup> PROM<br>FLASH | Mask-Programmed Programmable (PROM) |
| SRAM<br>DRAM      | FIFO<br>LIFO<br>Shift Register<br>CAM |                                       |                                     |

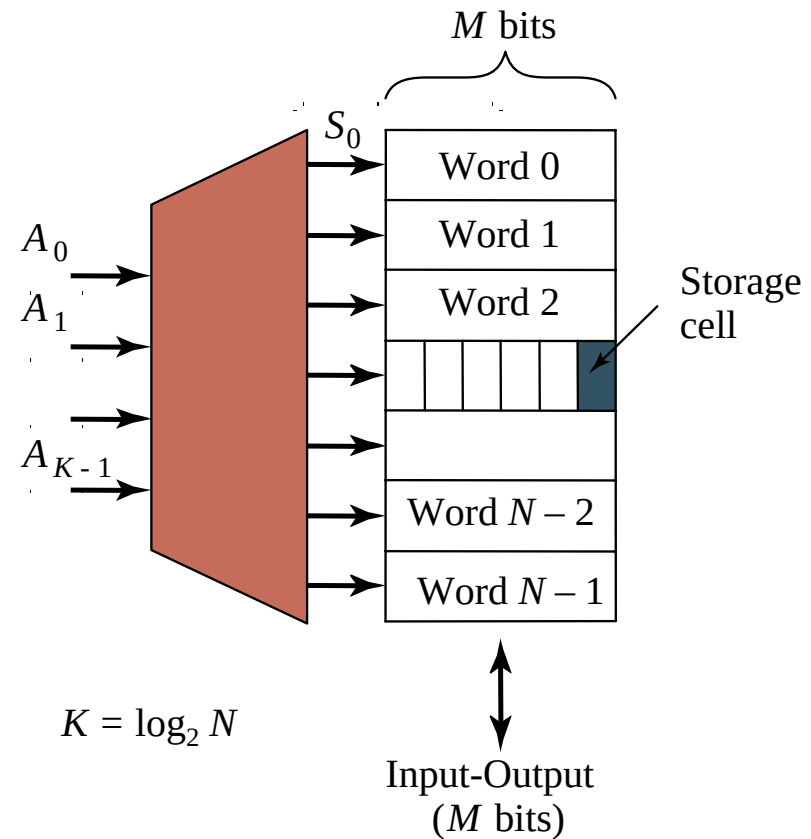
# Memory Timing: Definitions



# Memory Architecture: Decoders



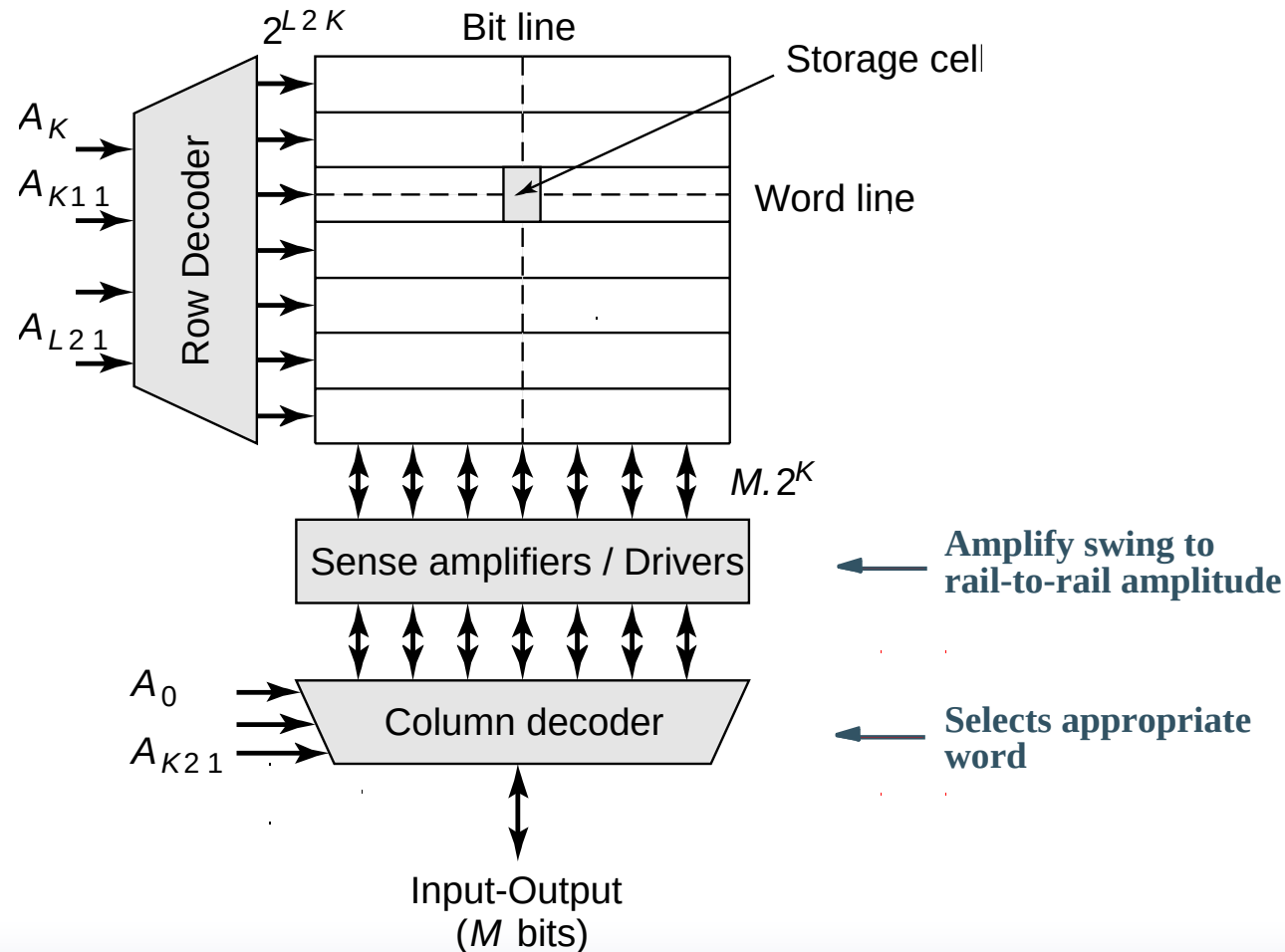
Intuitive architecture for  $N \times M$  memory  
Too many select signals:  
 $N$  words ==  $N$  select signals



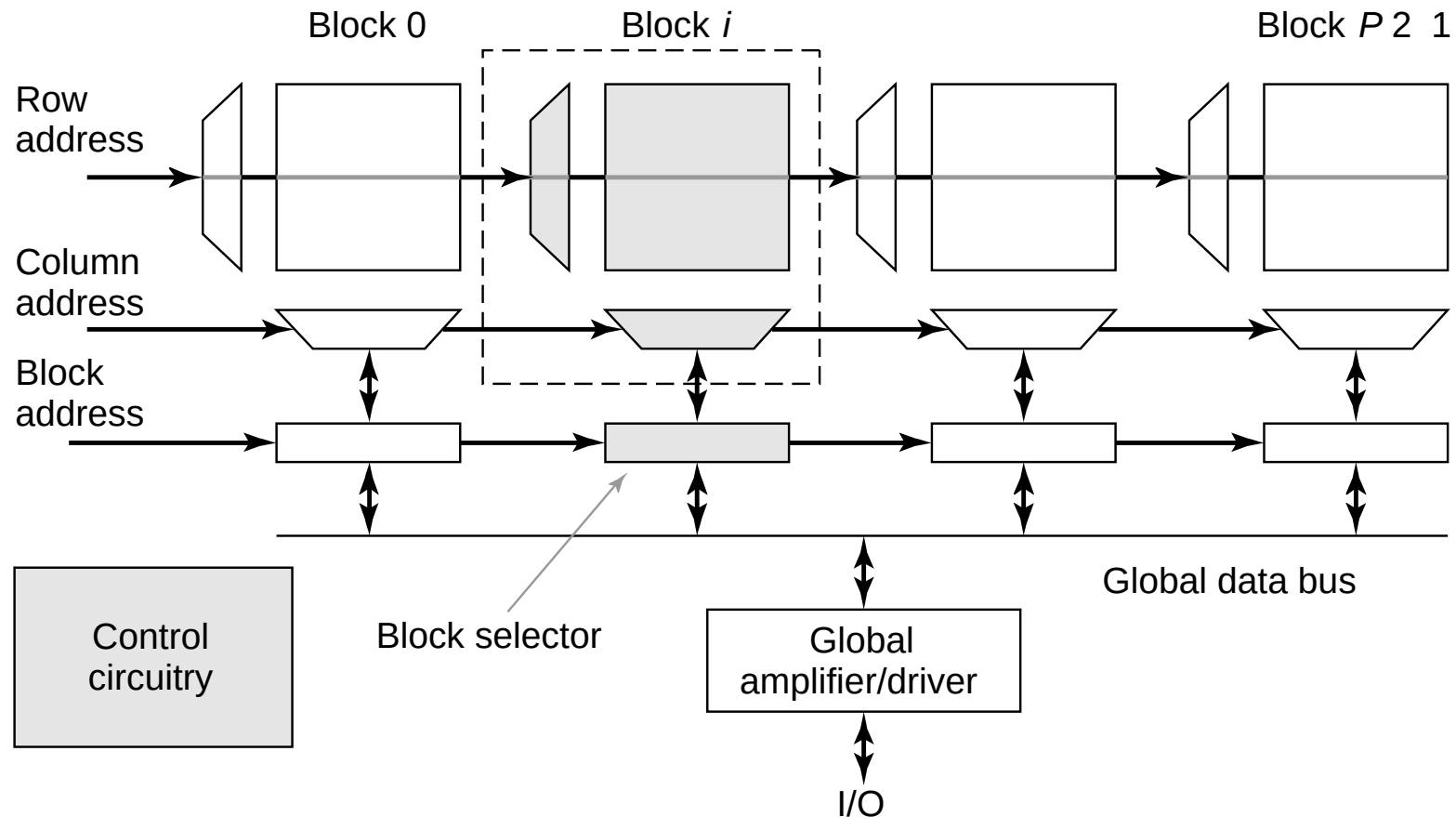
Decoder reduces the number of select signals  
 $K = \log_2 N$

# Array-Structured Memory Architecture

Problem: ASPECT RATIO or HEIGHT  $\gg$  WIDTH



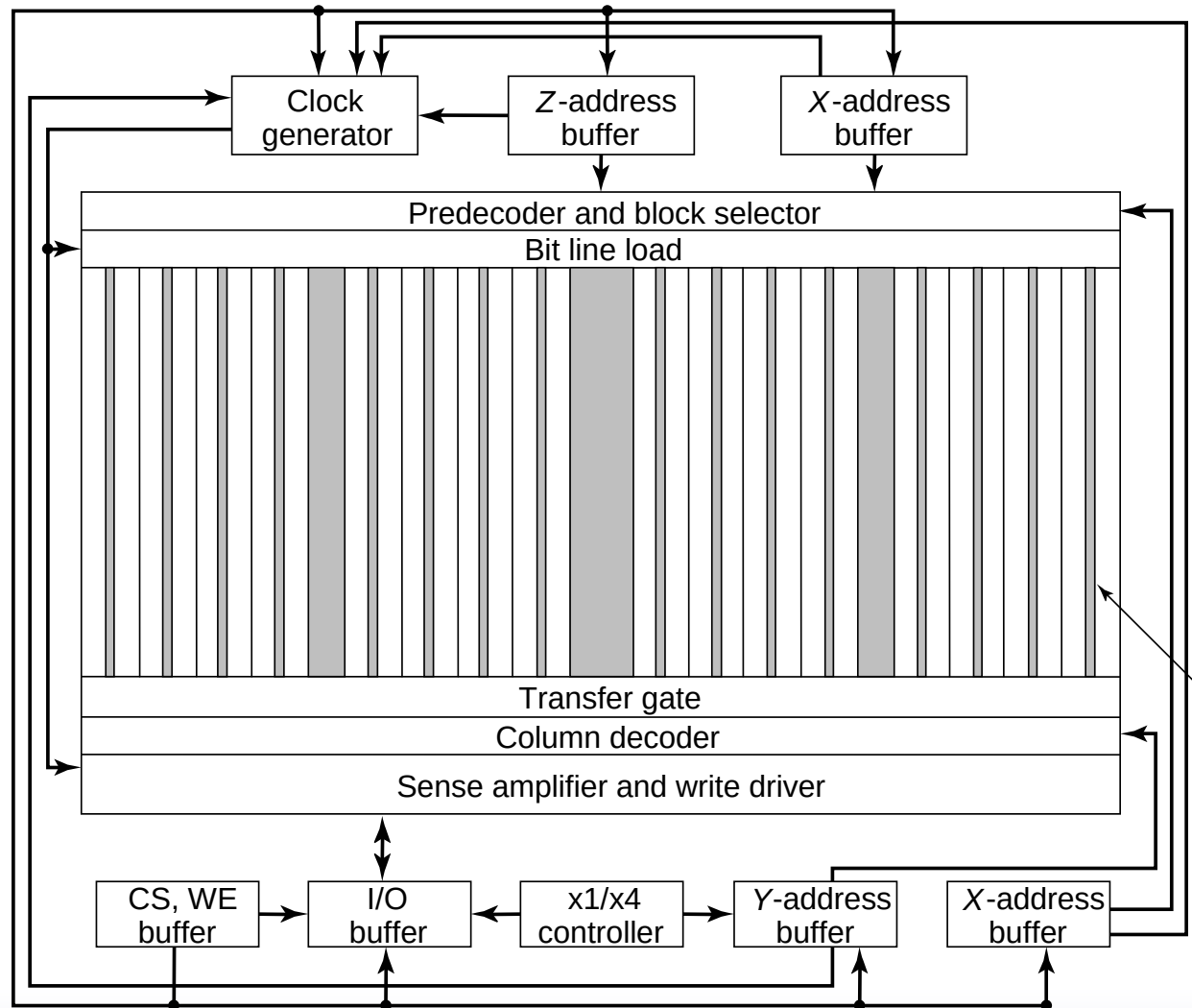
# Hierarchical Memory Architecture



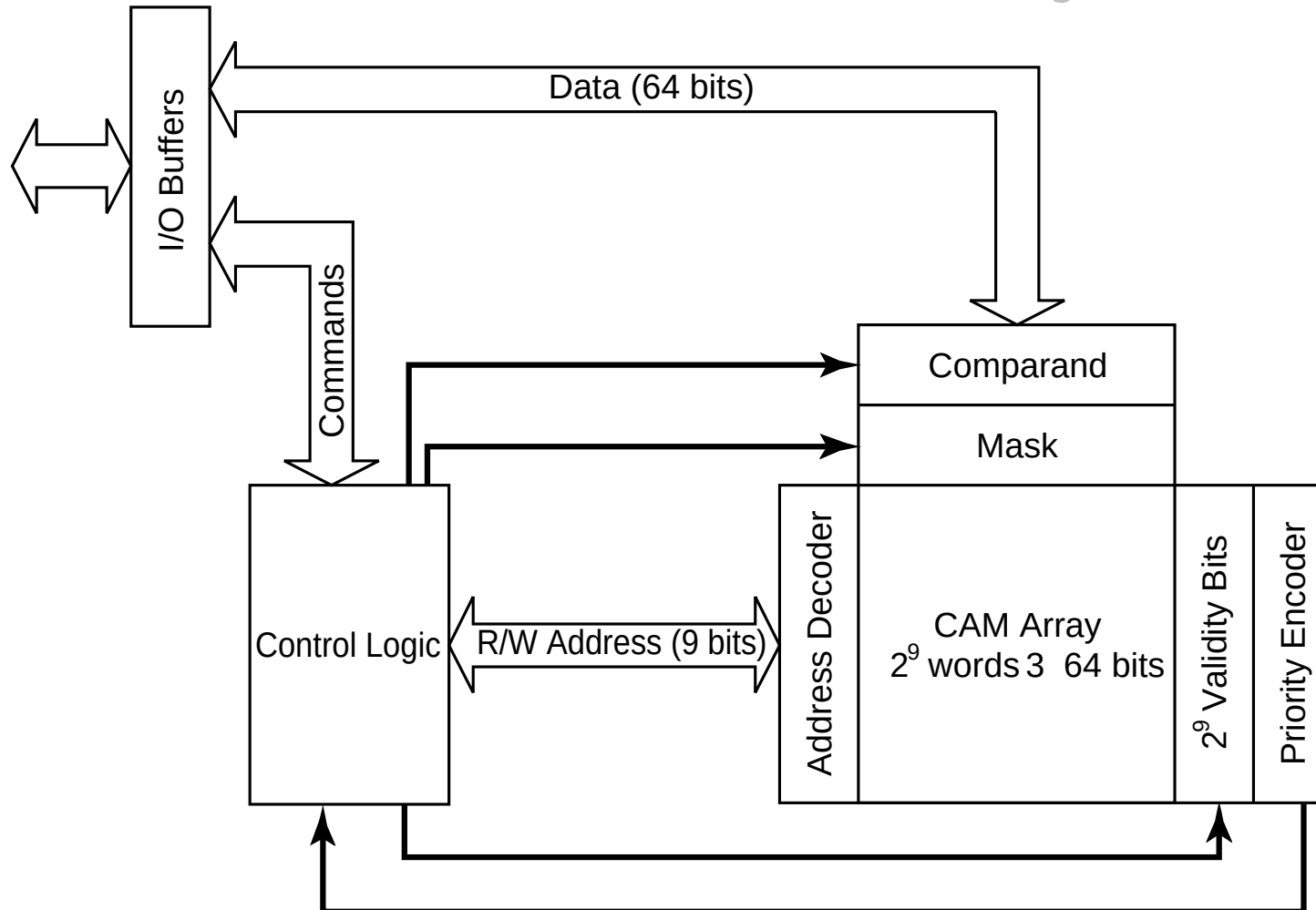
## Advantages:

1. Shorter wires within blocks
2. Block address activates only 1 block => power savings

# Block Diagram of 4 Mbit SRAM



# Contents-Addressable Memory



# ***Read-Write Memories (RAM)***

## **❑ STATIC (SRAM)**

**Data stored as long as supply is applied**

**Large (6 transistors/cell)**

**Fast**

**Differential**

## **❑ DYNAMIC (DRAM)**

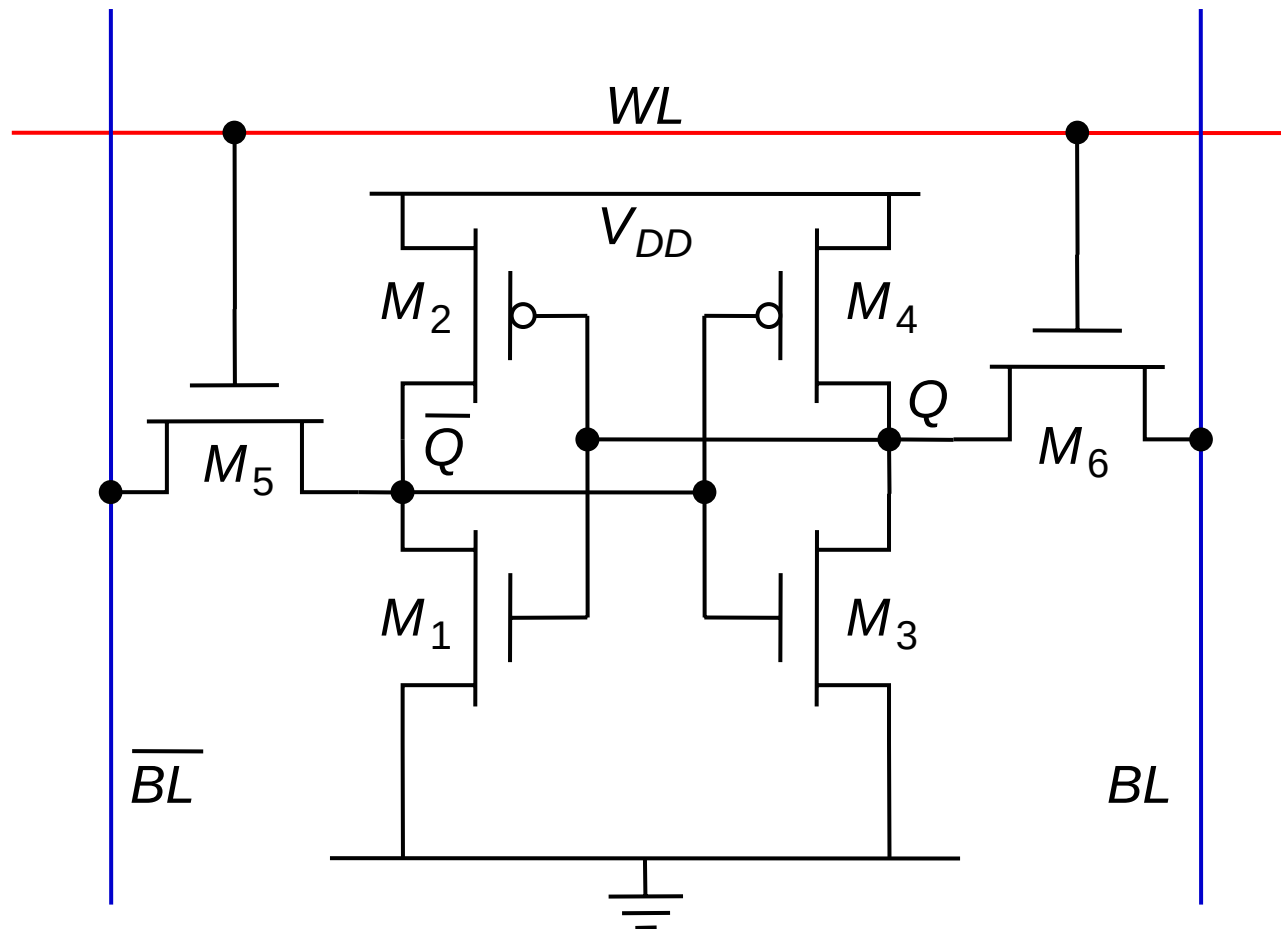
**Periodic refresh required**

**Small (1-3 transistors/cell)**

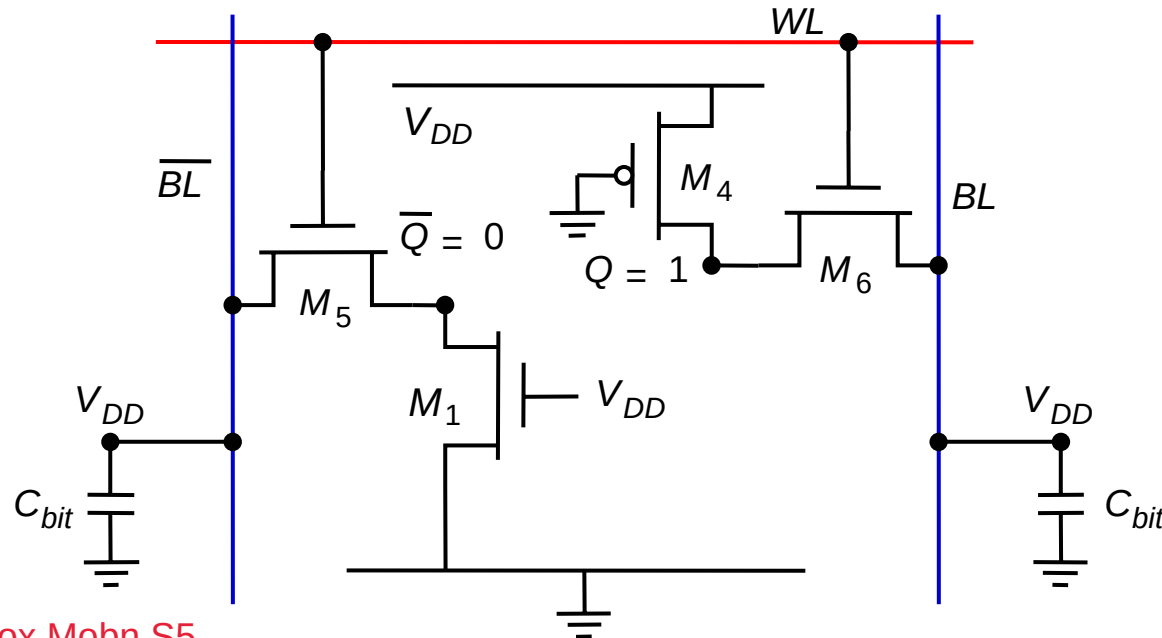
**Slower**

**Single Ended**

# 6-transistor CMOS SRAM Cell



# CMOS SRAM Analysis (Read)



Kn,M5=Cox Mobn S5

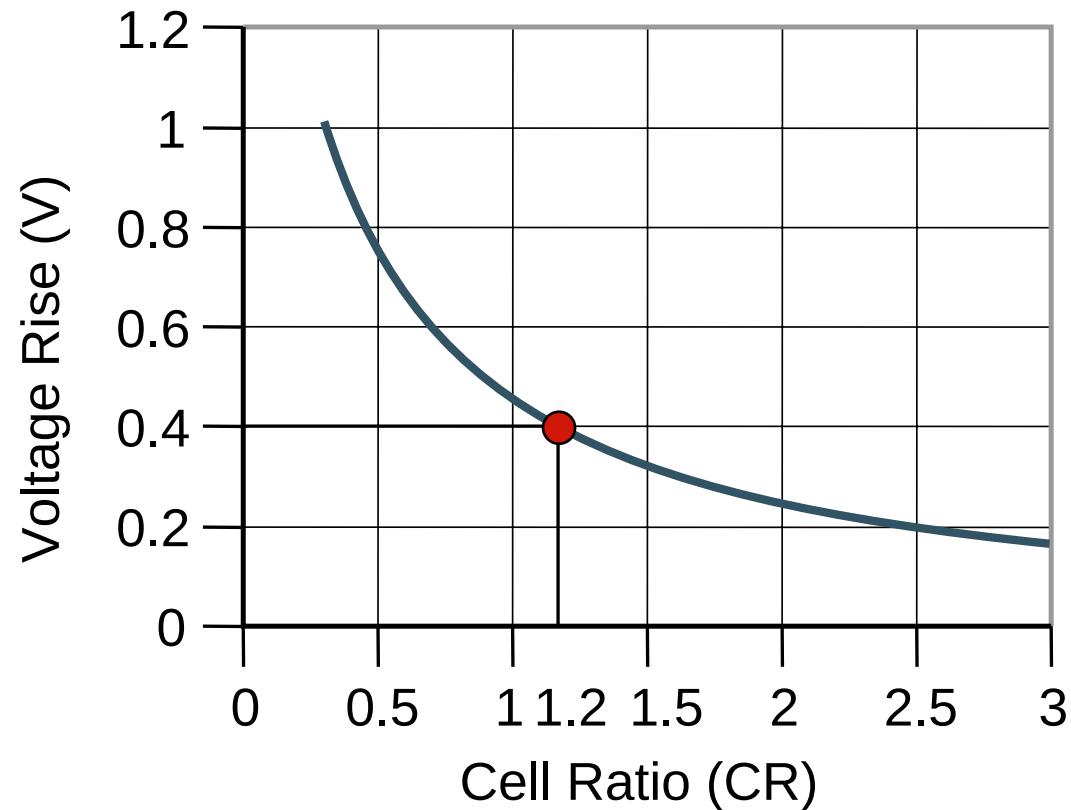
$$K_{n,M1} = C_{ox} M_{obn} S_1$$

$$k_{n,M5} \left( (V_{DD} - \Delta V - V_{Tn}) V_{DSATn} - \frac{V_{DSATn}^2}{2} \right) \stackrel{S1}{=} k_{n,M1} \left( (V_{DD} - V_{Tn}) \Delta V - \frac{\Delta V^2}{2} \right)$$

$$CR = S1/S5$$

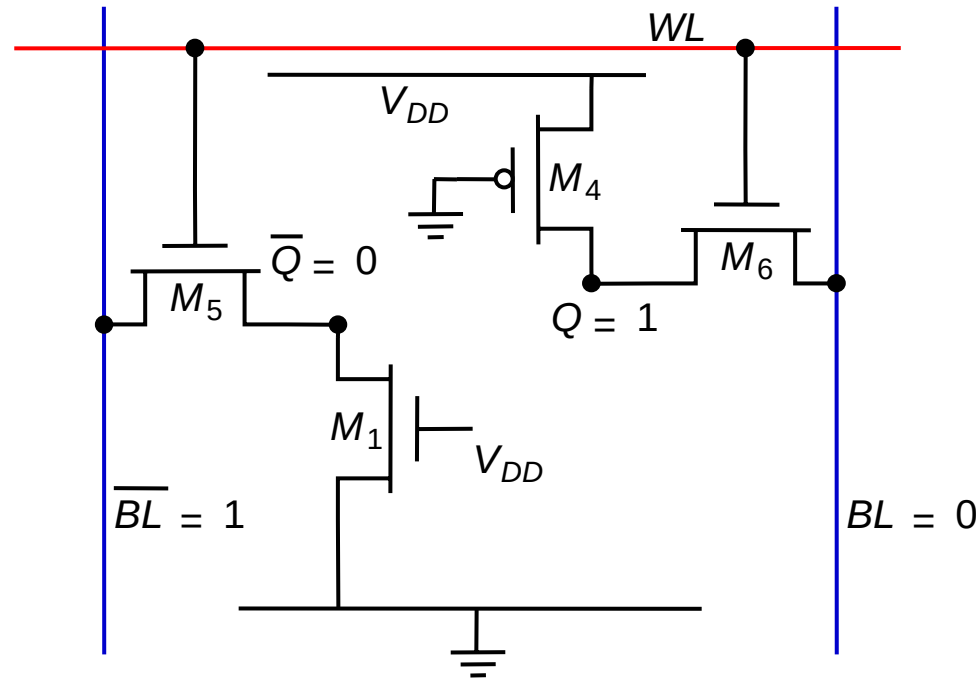
$$\Delta V = \frac{V_{DSATn} + CR(V_{DD} - V_{Tn}) - \sqrt{V_{DSATn}^2(1 + CR) + CR^2(V_{DD} - V_{Tn})^2}}{CR}$$

# CMOS SRAM Analysis (Read)



$$CR = \frac{W_1/L_1}{W_5/L_5}$$

# CMOS SRAM Analysis (Write)



$k_{n,M6} = C_{ox} \mu_{bn} S_6$

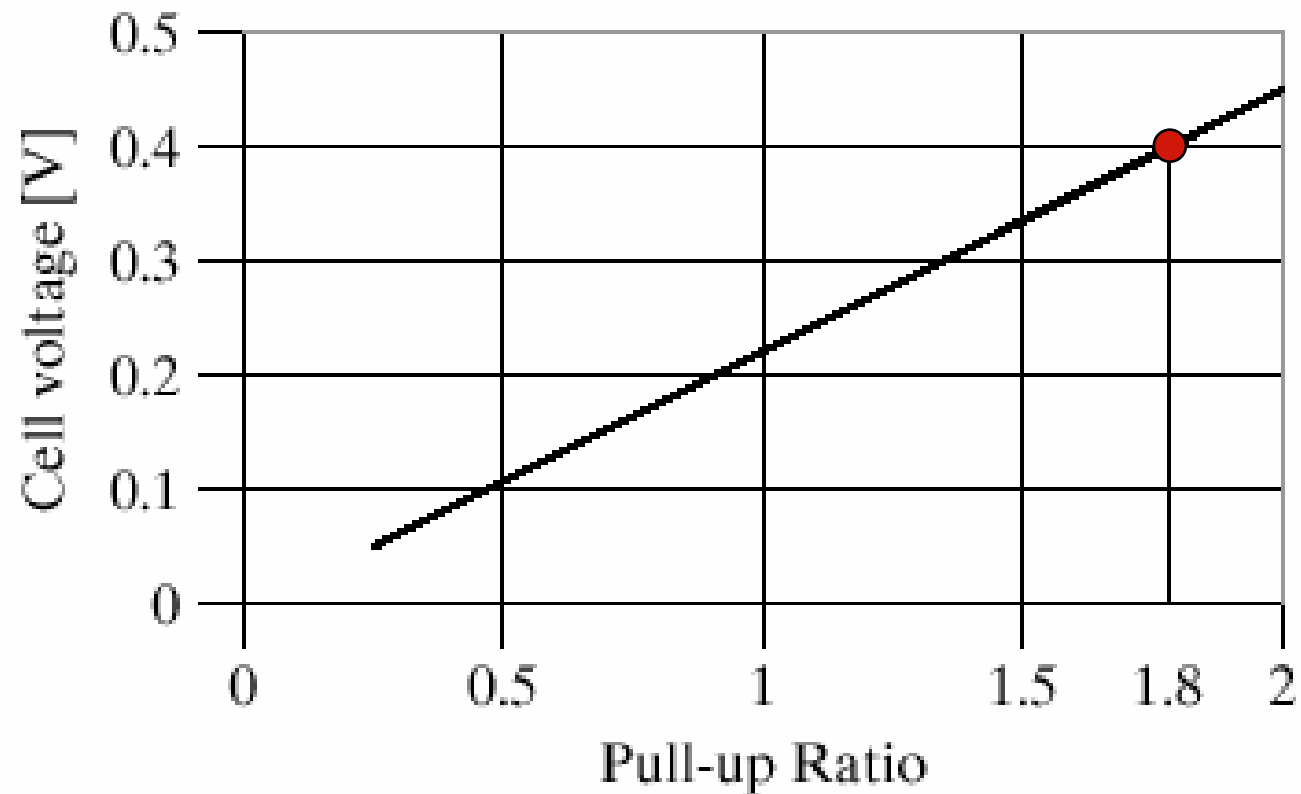
$k_{p,M4} = C_{ox} \mu_{bp} S_4$

$PR = S_4/S_6$

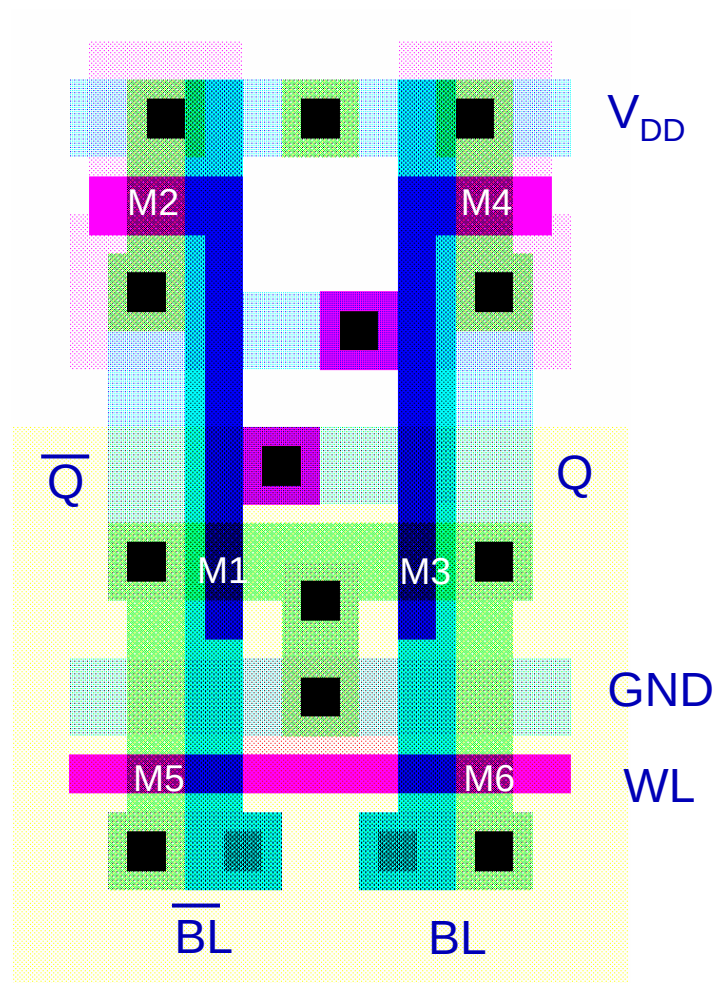
$$k_{n,M6} \left( (V_{DD} - V_{Tn}) V_Q - \frac{V_Q^2}{2} \right) = k_{p,M4} \left( (V_{DD} - |V_{Tp}|) V_{DSATp} - \frac{V_{DSATp}^2}{2} \right)$$

$$V_Q = V_{DD} - V_{Tn} - \sqrt{(V_{DD} - V_{Tn})^2 - 2 \frac{\mu_p}{\mu_n} PR \left( (V_{DD} - |V_{Tp}|) V_{DSATp} - \frac{V_{DSATp}^2}{2} \right)},$$

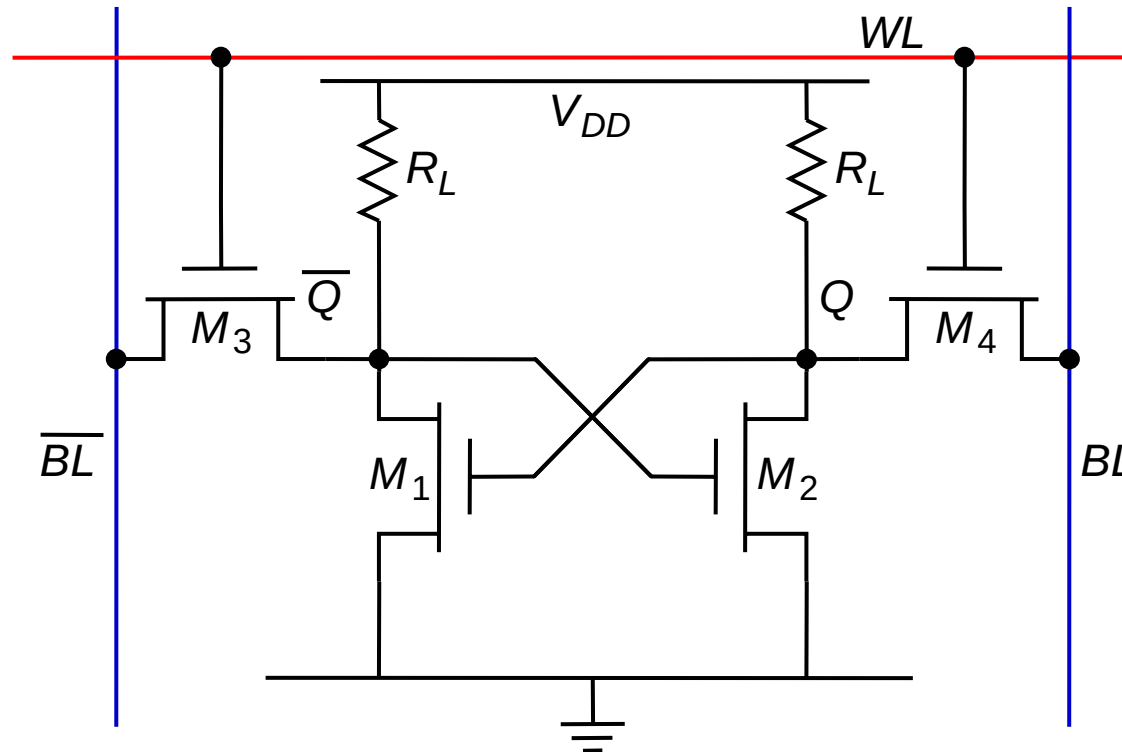
# CMOS SRAM Analysis (Write)



# 6T-SRAM — Layout



# Resistance-load SRAM Cell



Static power dissipation -- Want  $R_L$  large  
Bit lines precharged to  $V_{DD}$  to address  $t_p$  problem

# SRAM Characteristics

**Table 12-2** Comparison of CMOS SRAM cells used in 1-Mbit memory (from [Takada91])

|                            | <b>Complementary CMOS</b>                         | <b>Resistive Load</b>                             | <b>TFT Cell</b>                                   |
|----------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Number of transistors      | 6                                                 | 4                                                 | 4 (+2 TFT)                                        |
| Cell size                  | 58.2 $\mu\text{m}^2$<br>(0.7- $\mu\text{m}$ rule) | 40.8 $\mu\text{m}^2$<br>(0.7- $\mu\text{m}$ rule) | 41.1 $\mu\text{m}^2$<br>(0.8- $\mu\text{m}$ rule) |
| Standby current (per cell) | $10^{-15}$ A                                      | $10^{-12}$ A                                      | $10^{-13}$ A                                      |

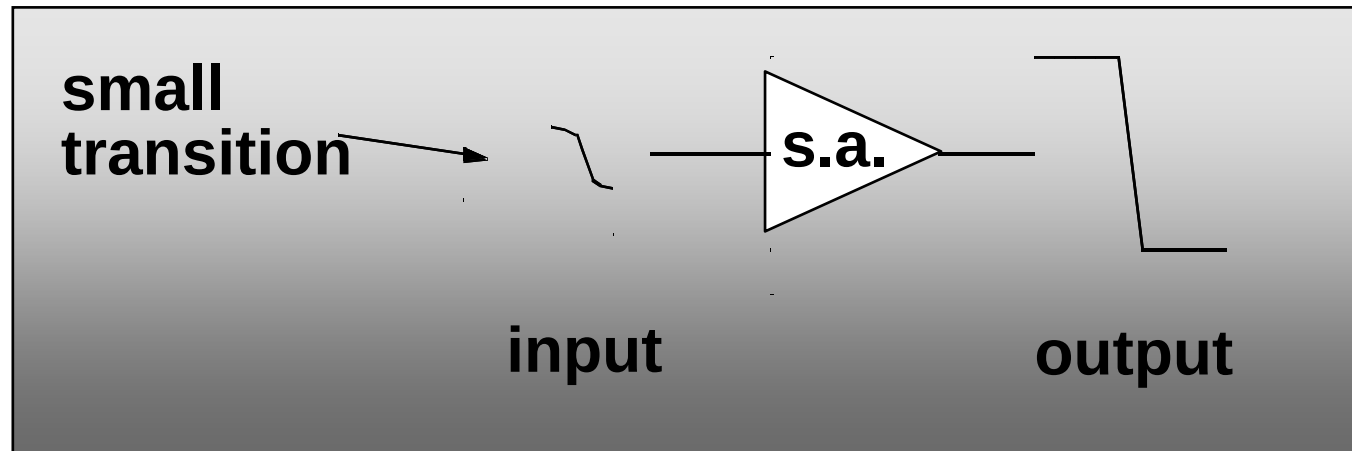
# Sense Amplifiers

$$t_p = \frac{C \cdot \Delta V}{I_{av}}$$

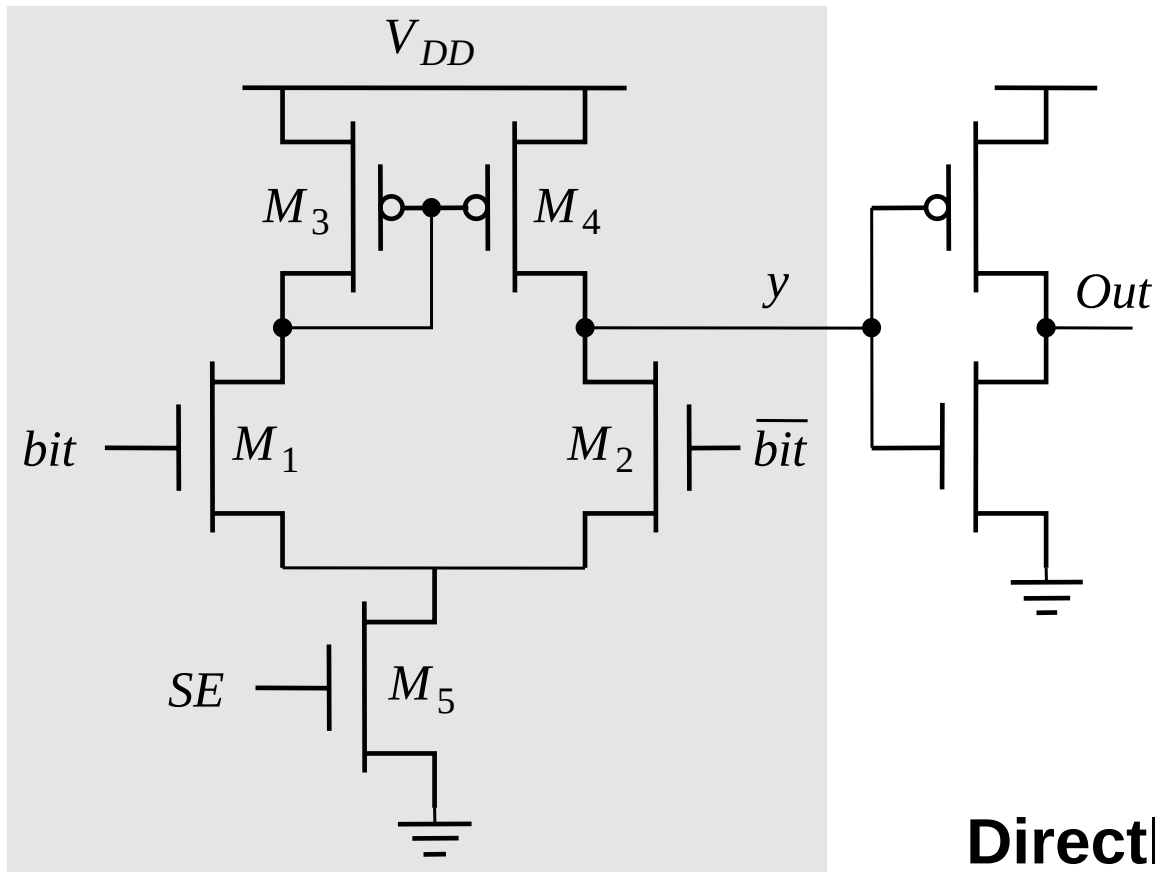
Annotations:

- make  $\Delta V$  as small as possible (points to  $\Delta V$ )
- large (points to  $C$ )
- small (points to  $I_{av}$ )

**Idea: Use Sense Amplifier**

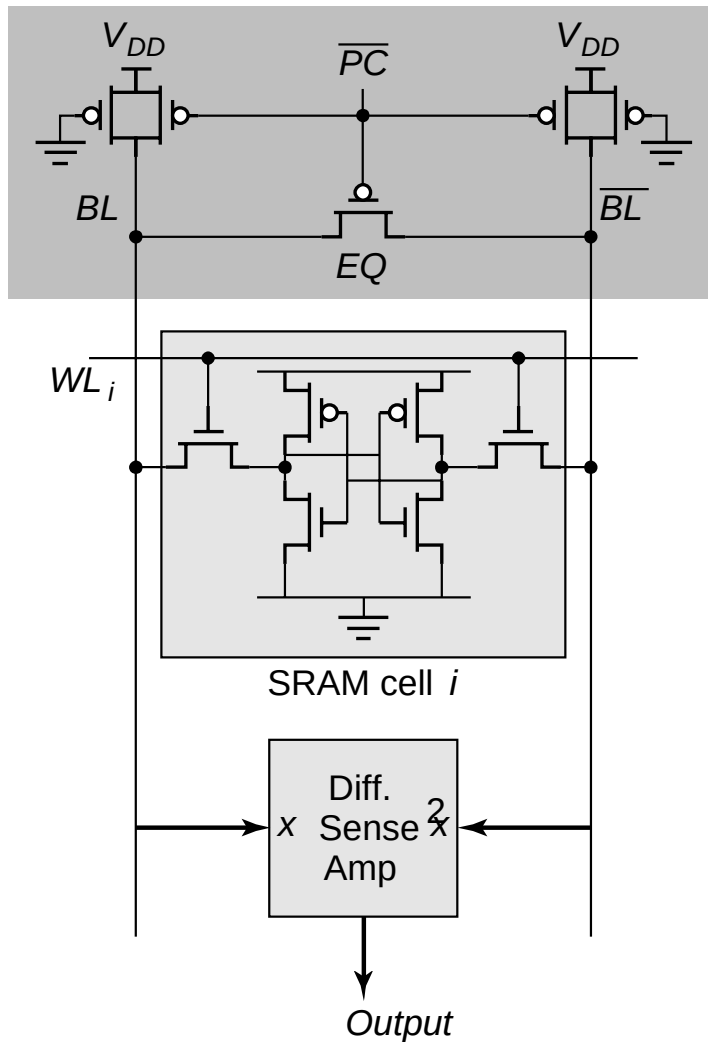


# Differential Sense Amplifier

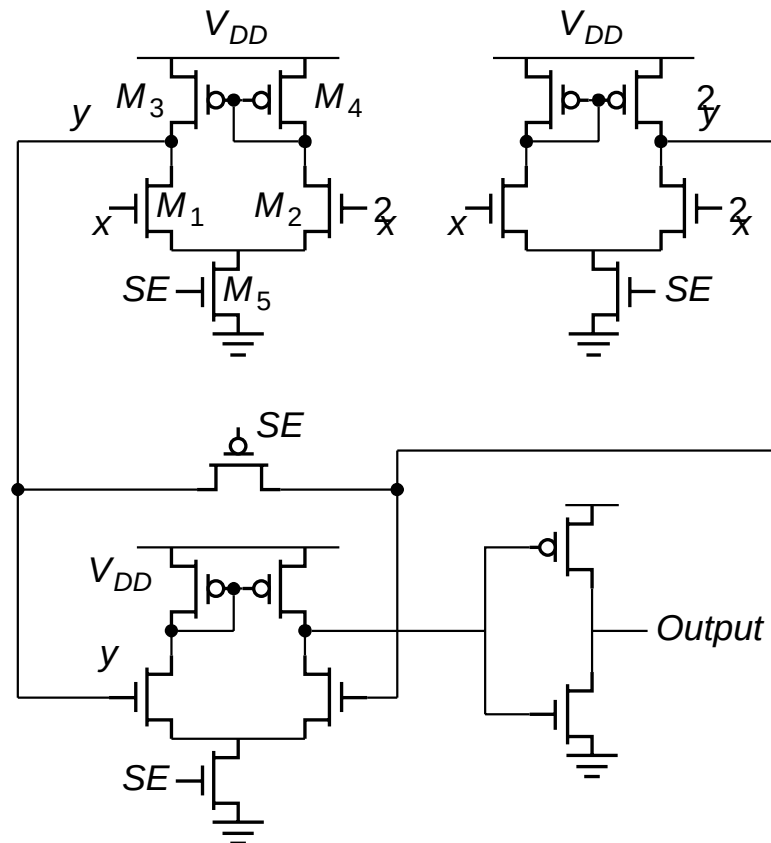


**Directly applicable to  
SRAMs**

# Differential Sensing — SRAM

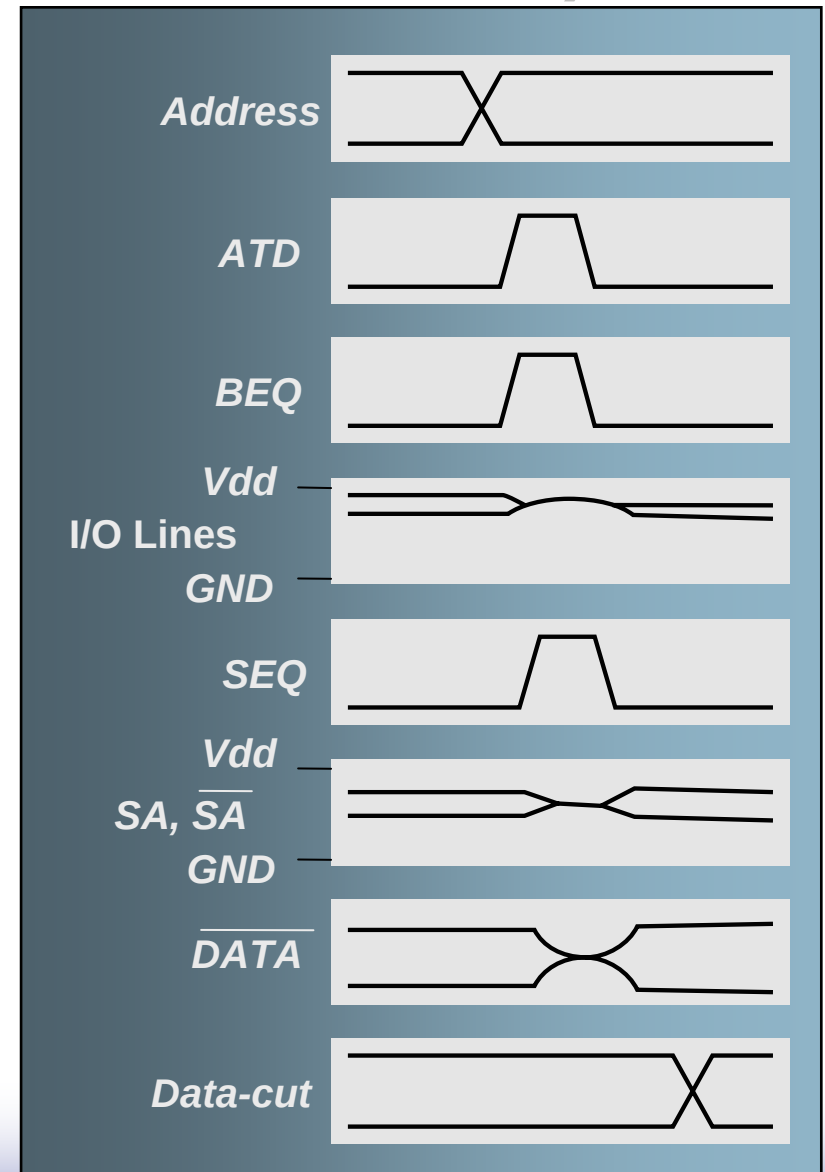
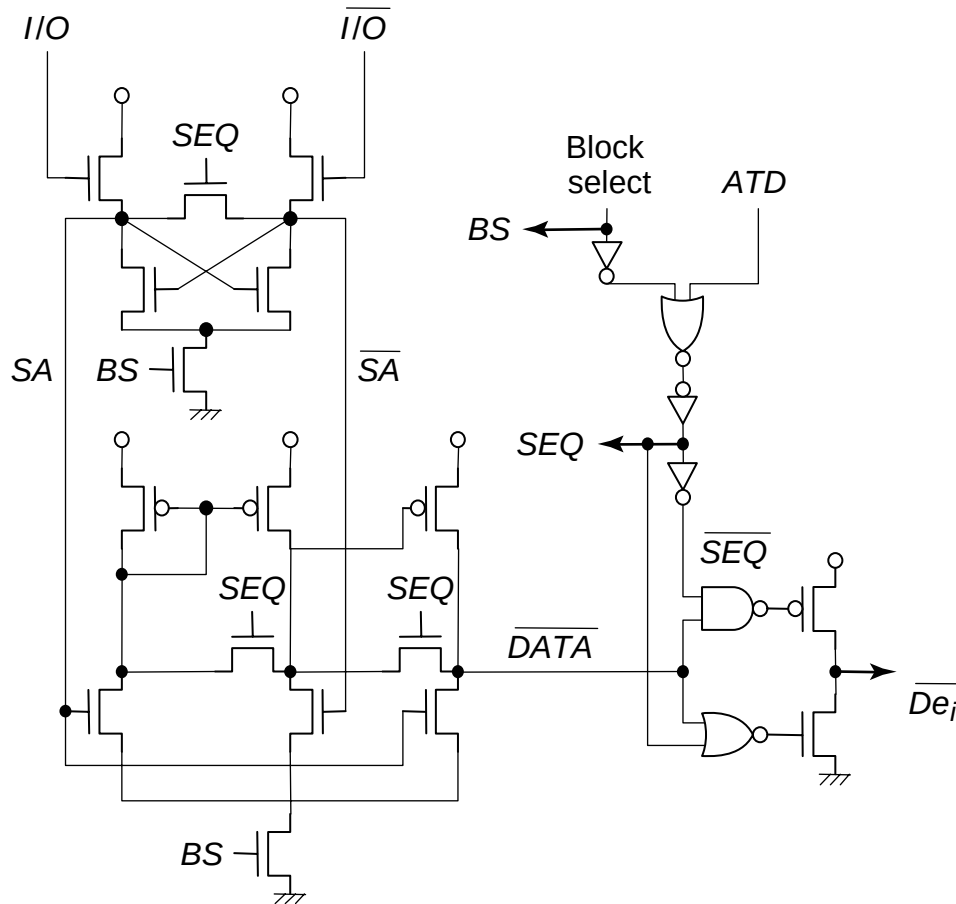


(a) SRAM sensing scheme

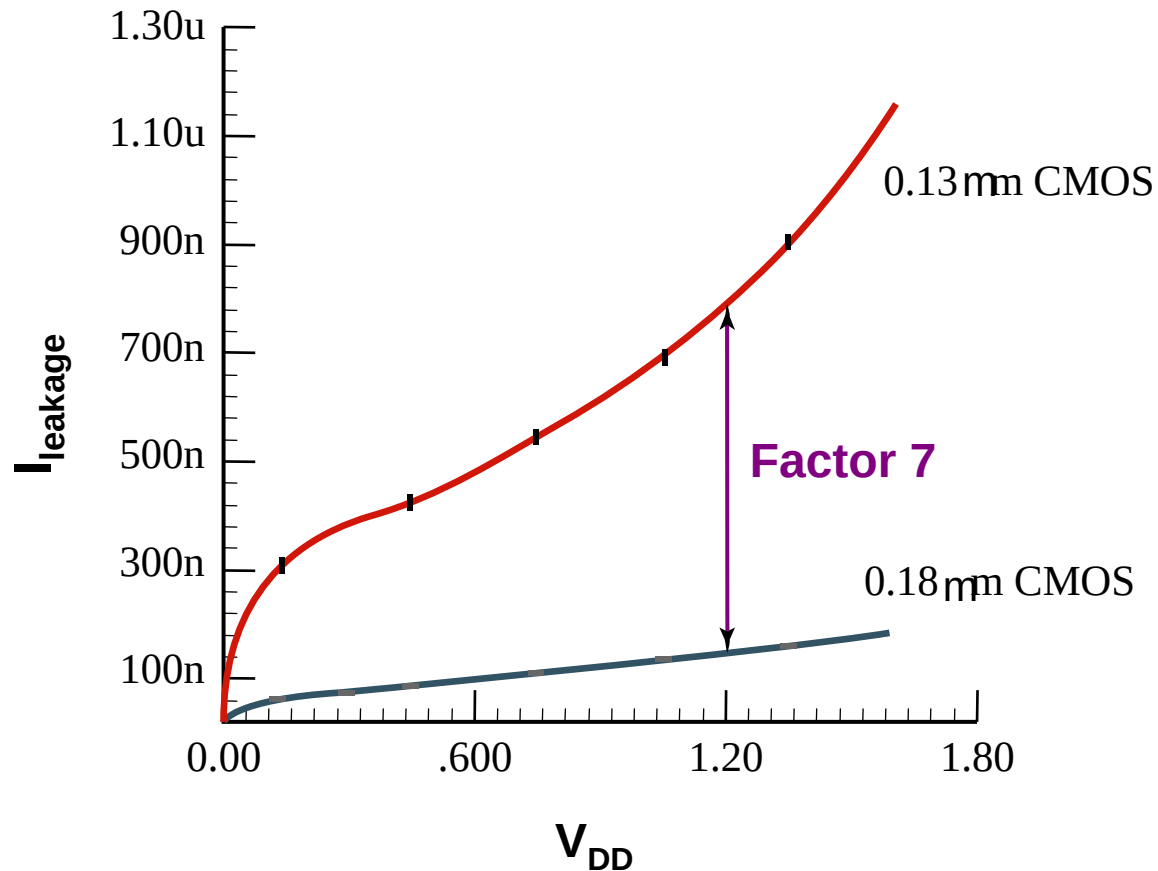


(b) two stage differential amplifier

# Sense Amplifier (and Waveforms)

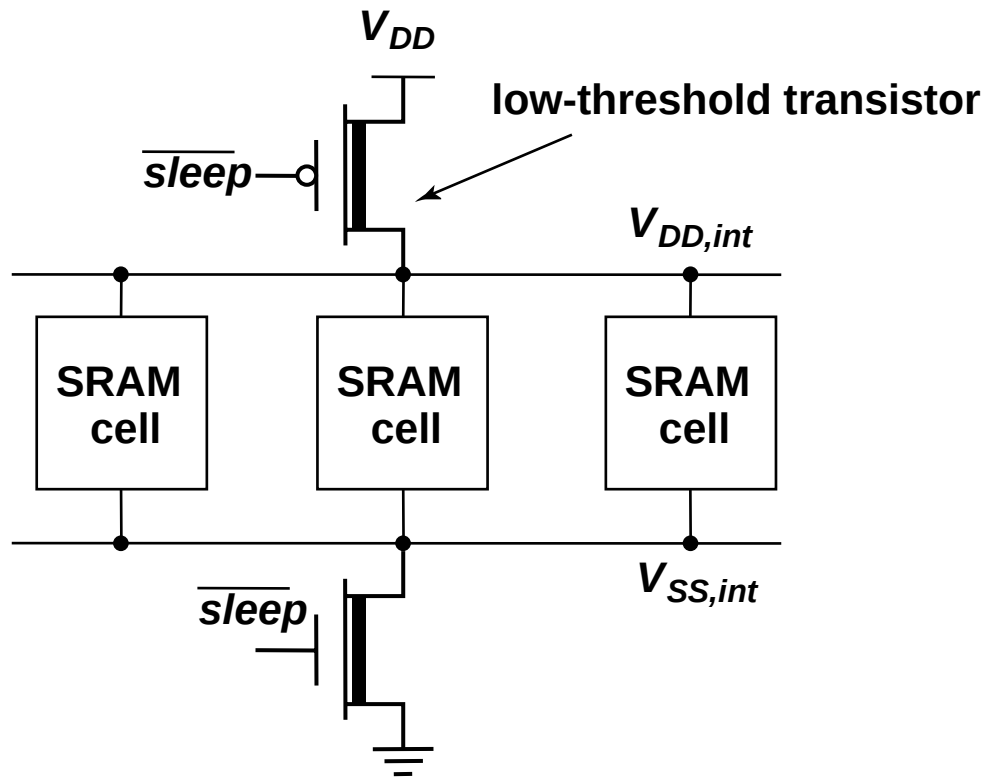


# Data Retention in SRAM

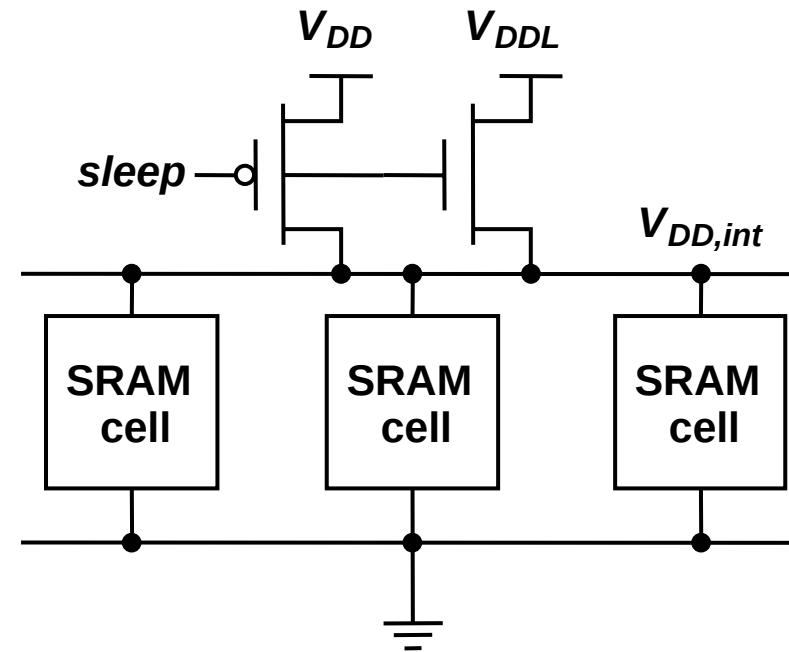


***SRAM leakage increases with technology scaling***

# Suppressing Leakage in SRAM

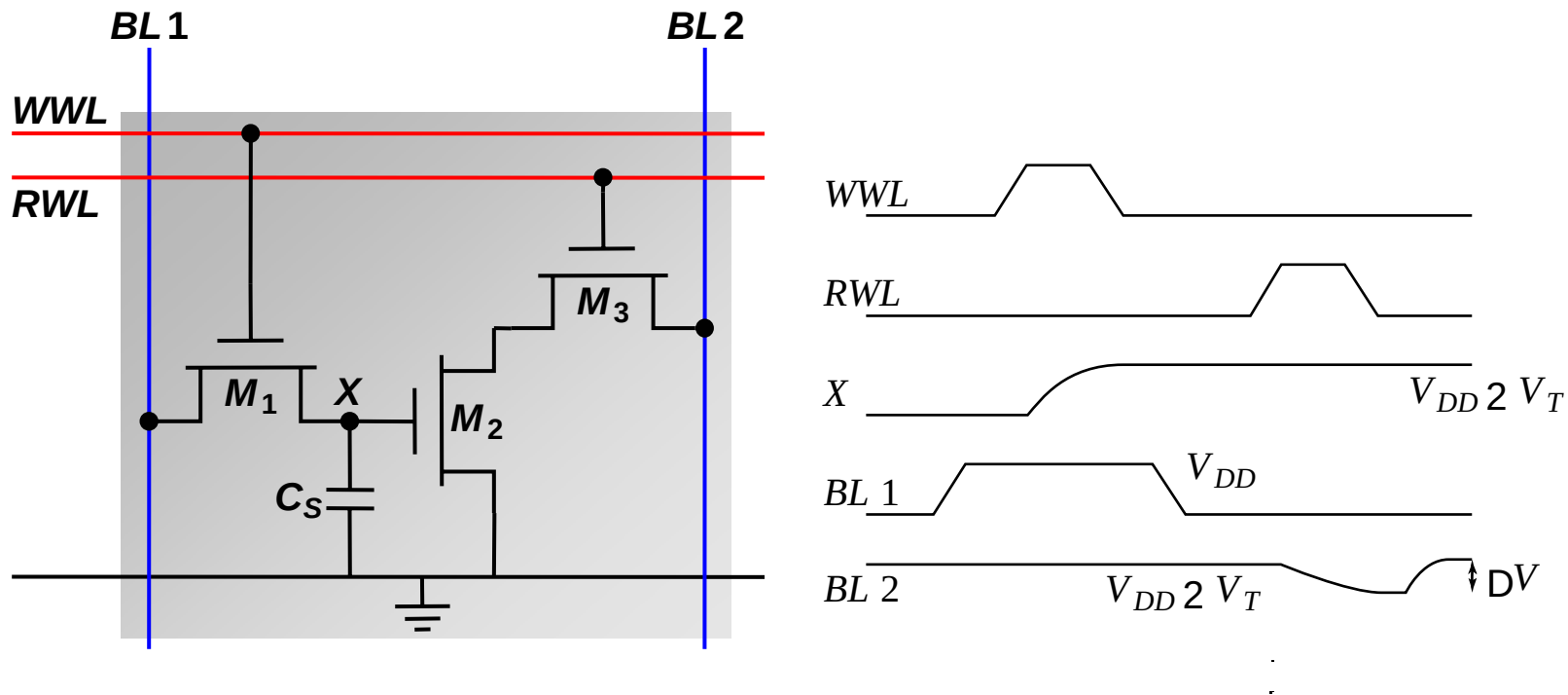


*Inserting Extra Resistance*



*Reducing the supply voltage*

# 3-Transistor DRAM Cell

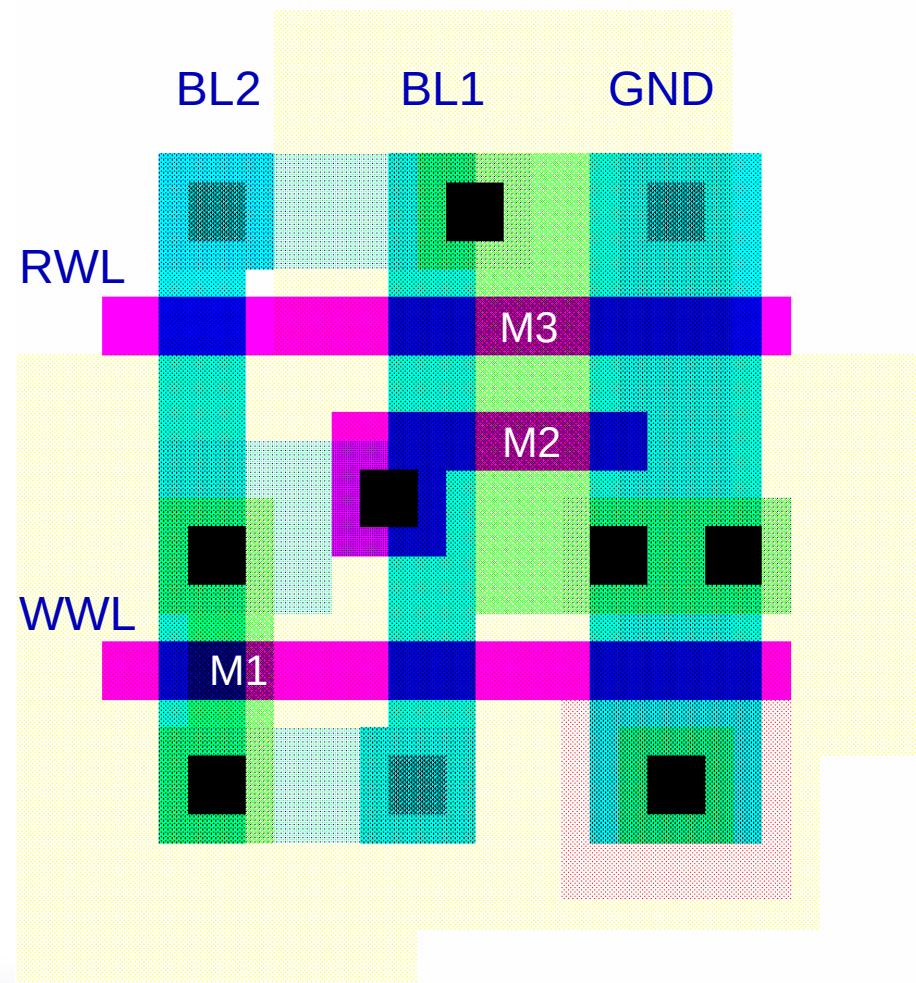


No constraints on device ratios

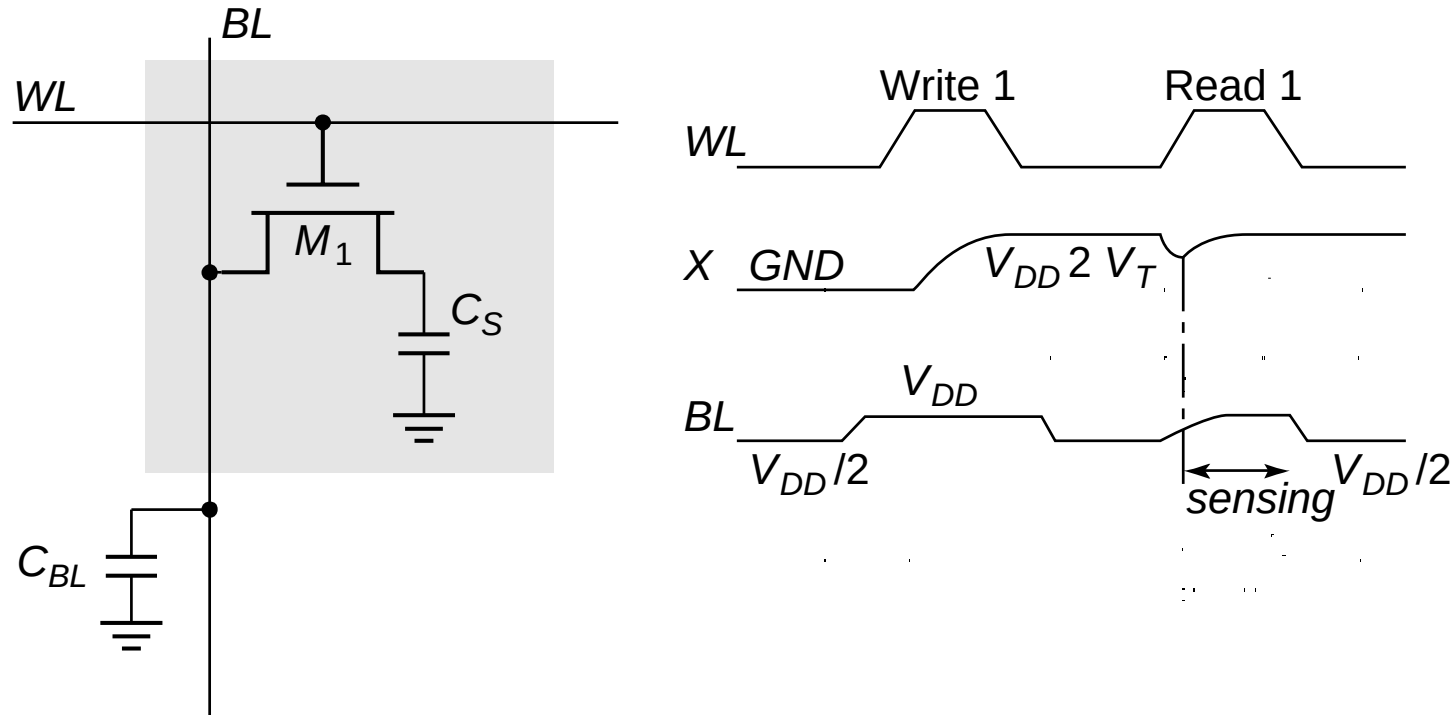
Reads are non-destructive

Value stored at node X when writing a "1" =  $V_{WWL} - V_{Tn}$

# 3T-DRAM — Layout



# 1-Transistor DRAM Cell



**Write:**  $C_S$  is charged or discharged by asserting WL and BL.

**Read:** Charge redistribution takes places between bit line and storage capacitance

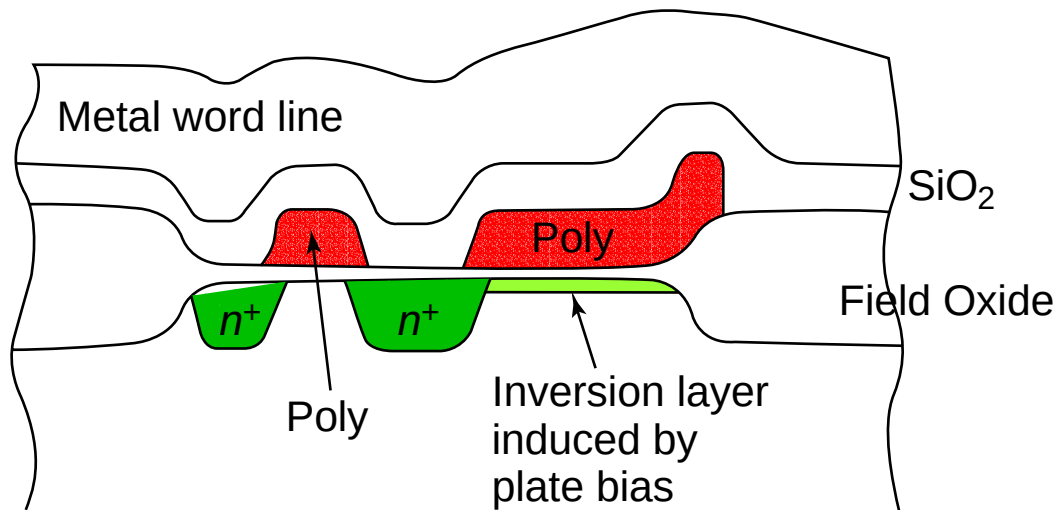
$$\Delta V = V_{BL} - V_{PRE} = V_{BIT} - V_{PRE} \frac{C_S}{C_S + C_{BL}}$$

**Voltage swing is small; typically around 250 mV.**

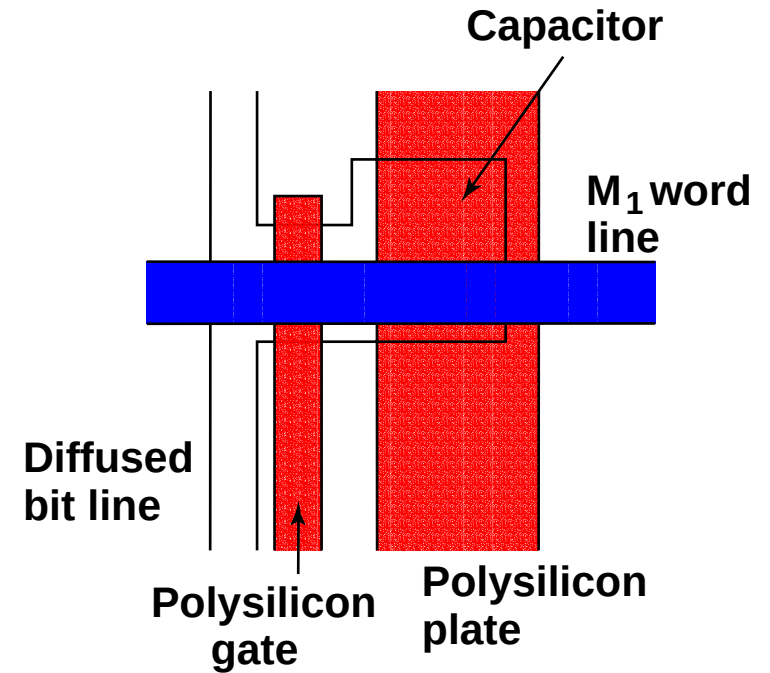
# ***DRAM Cell Observations***

- ❑ 1T DRAM requires a sense amplifier for each bit line, due to charge redistribution read-out.
- ❑ DRAM memory cells are single ended in contrast to SRAM cells.
- ❑ The read-out of the 1T DRAM cell is destructive; read and refresh operations are necessary for correct operation.
- ❑ Unlike 3T cell, 1T cell requires presence of an extra capacitance that must be explicitly included in the design.
- ❑ When writing a “1” into a DRAM cell, a threshold voltage is lost. This charge loss can be circumvented by bootstrapping the word lines to a higher value than  $V_{DD}$

# 1-T DRAM Cell



**Cross-section**



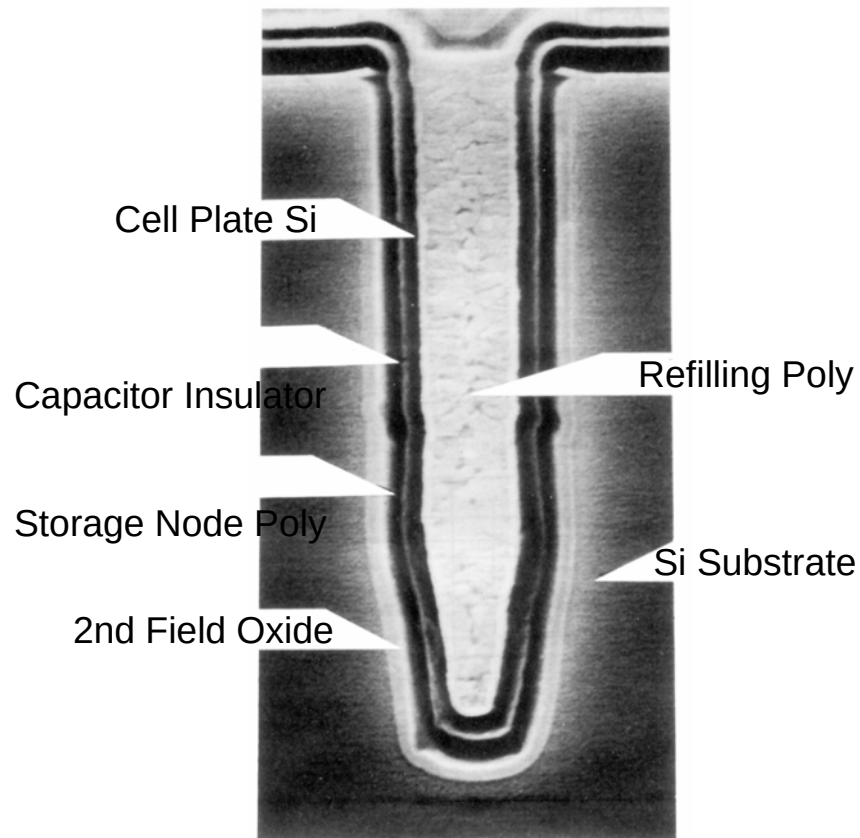
**Layout**

**Uses Polysilicon-Diffusion Capacitance**  
**Expensive in Area**

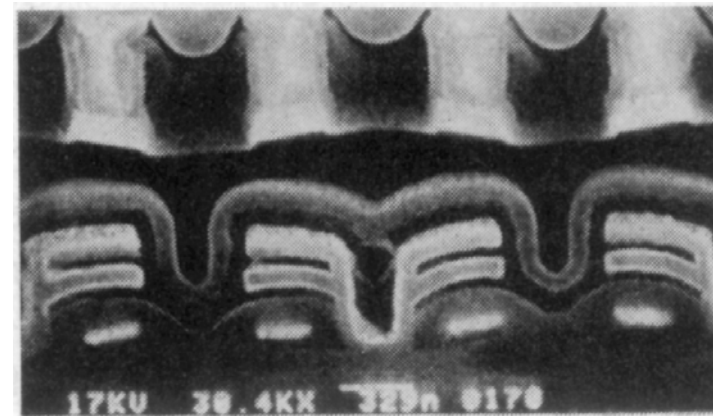
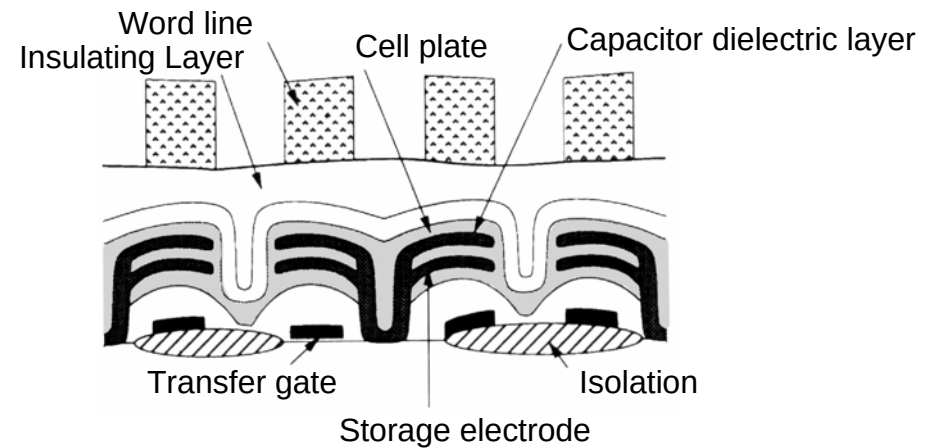
## *SEM of poly-diffusion capacitor 1T-DRAM*



# Advanced 1T DRAM Cells

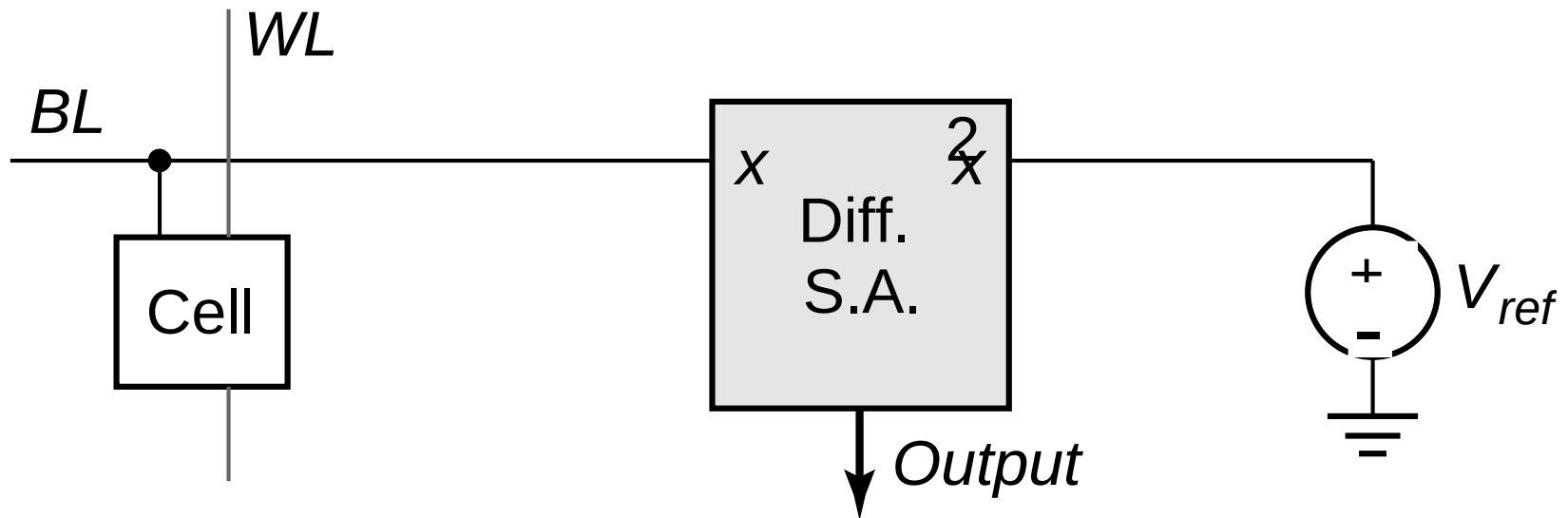


**Trench Cell**



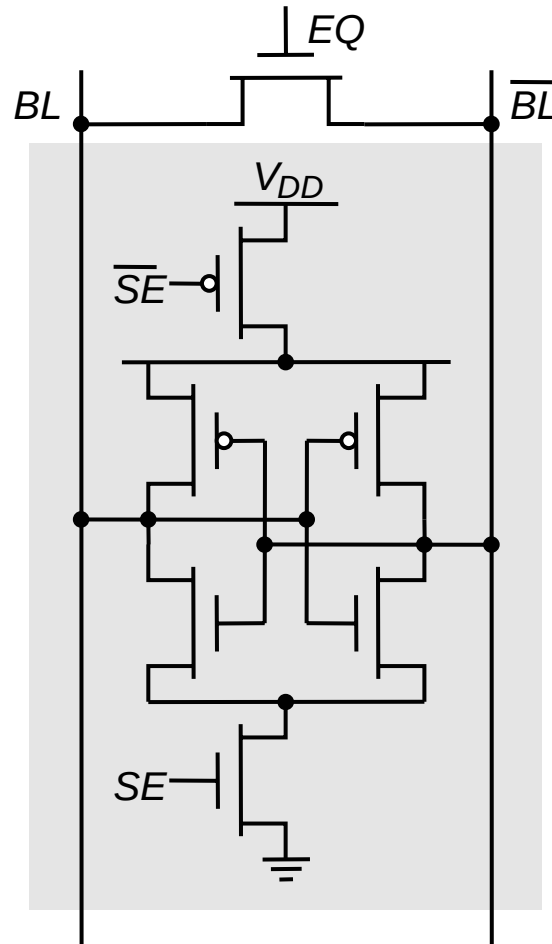
**Stacked-capacitor Cell**

# Single-to-Differential Conversion



**How to make a good  $V_{ref}$ ?**

# Latch-Based Sense Amplifier (DRAM)

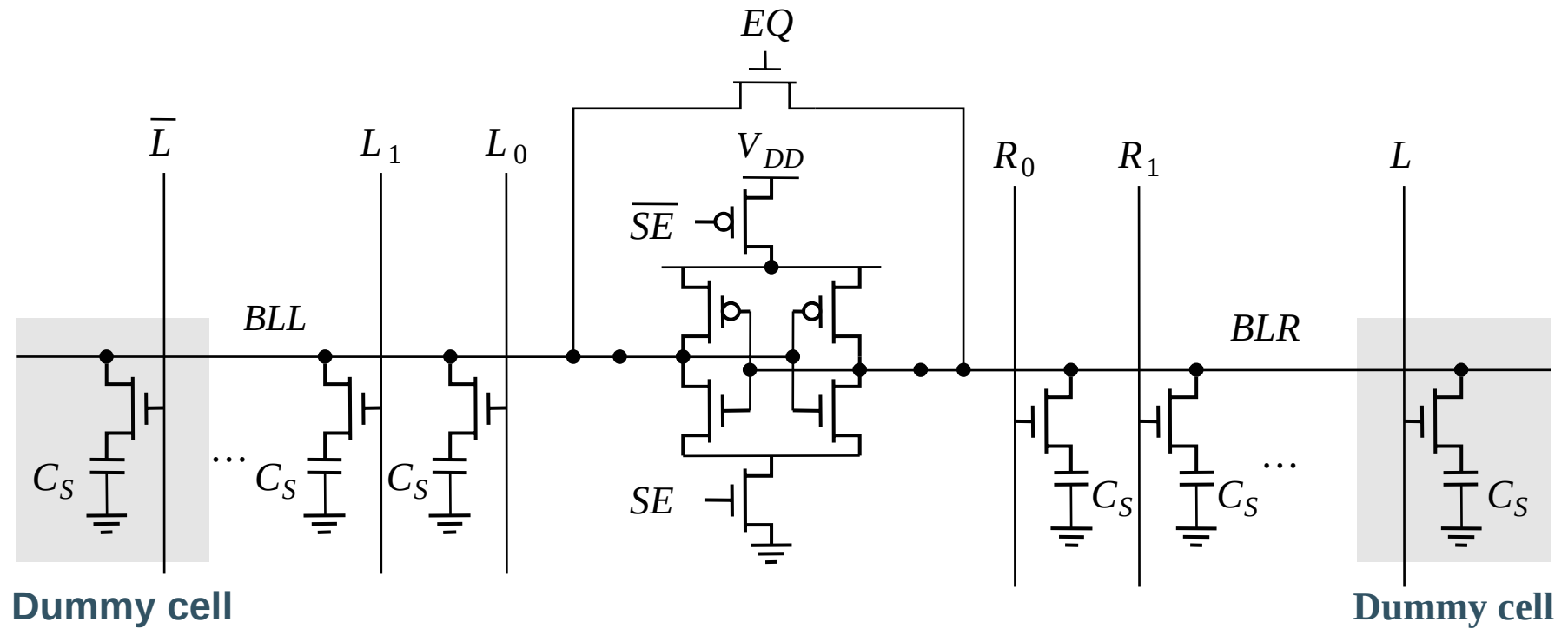


## Initialized in its meta-stable point with EQ

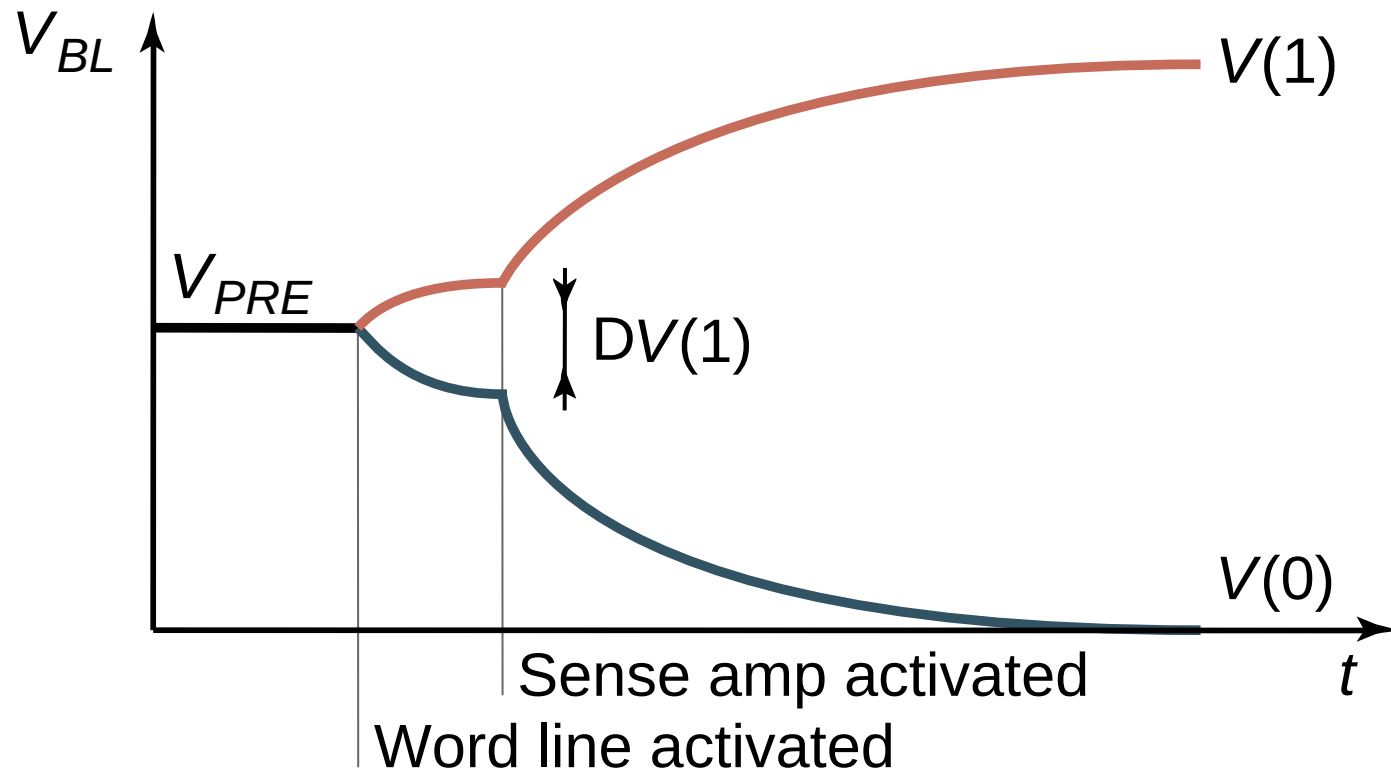
Once adequate voltage gap created, sense amp enabled with SE

Positive feedback quickly forces output to a stable operating point.

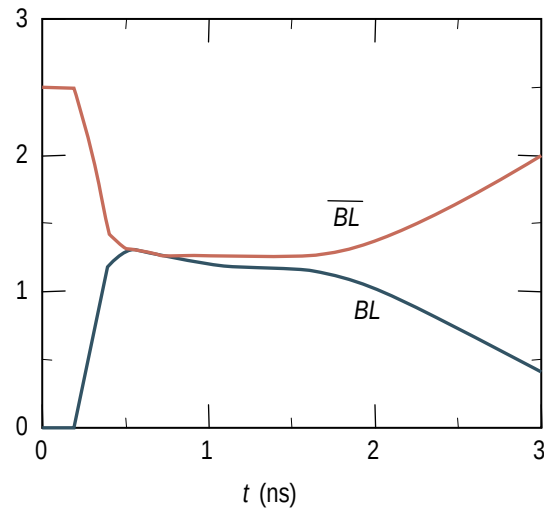
# Open bitline architecture with dummy cells



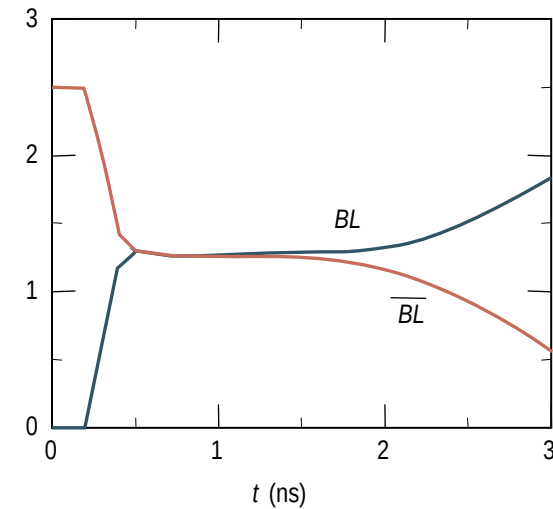
# Sense Amp Operation



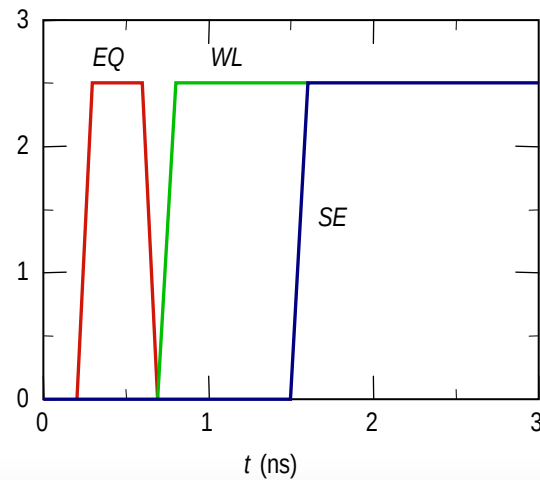
# DRAM Read Process with Dummy Cell



reading 0



reading 1



control signals

# *Periphery*

- ☐ **Decoders**
- ☐ **Sense Amplifiers**
- ☐ **Input/Output Buffers**
- ☐ **Control / Timing Circuitry**

# Row Decoders

Collection of  $2^M$  complex logic gates  
Organized in regular and dense fashion

## (N)AND Decoder

$$WL_0 = A_0 A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 A_9$$

$$WL_{511} = \bar{A}_0 A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 A_9$$

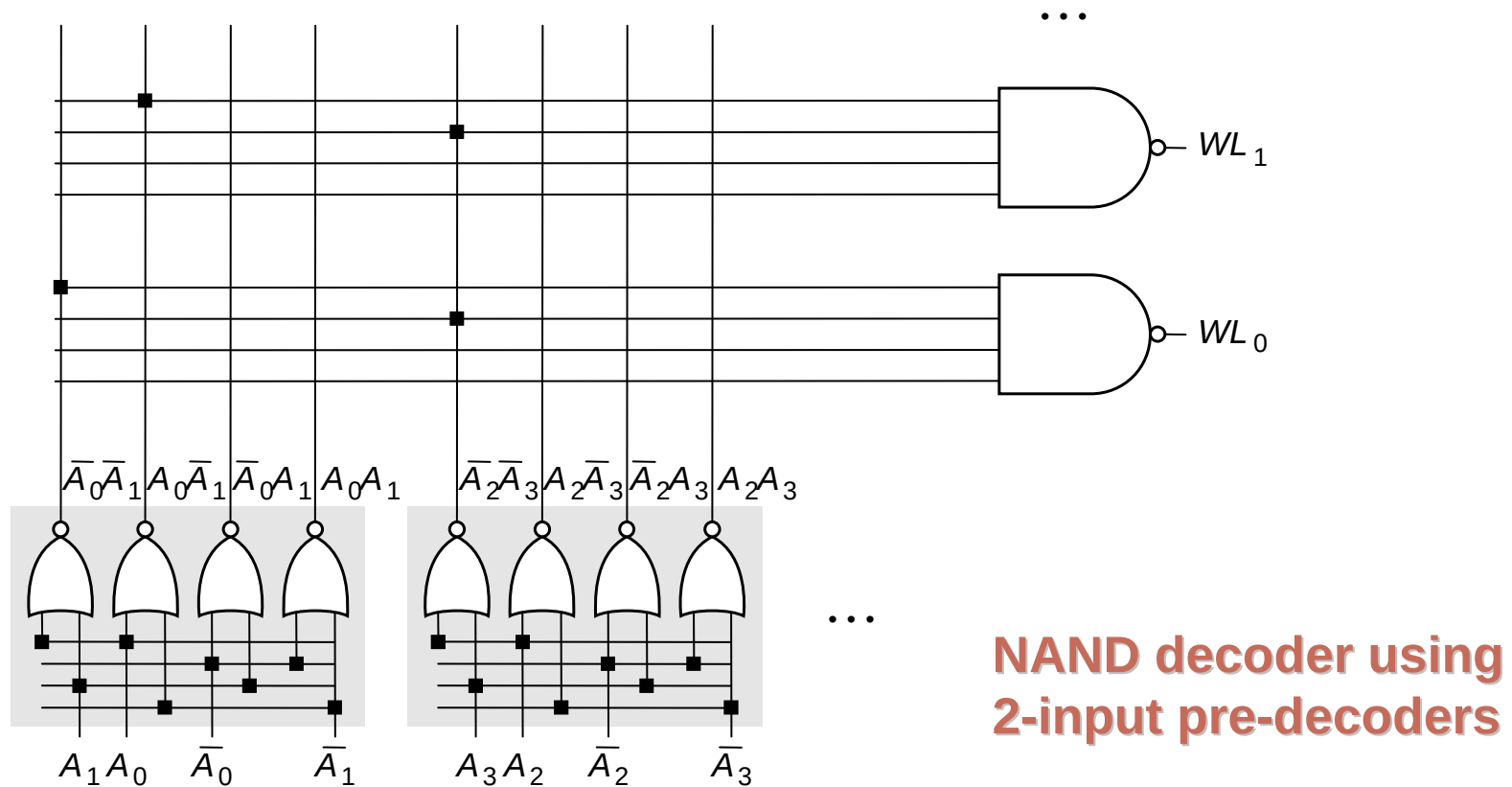
## NOR Decoder

$$WL_0 = \overline{A_0 + A_1 + A_2 + A_3 + A_4 + A_5 + A_6 + A_7 + A_8 + A_9}$$

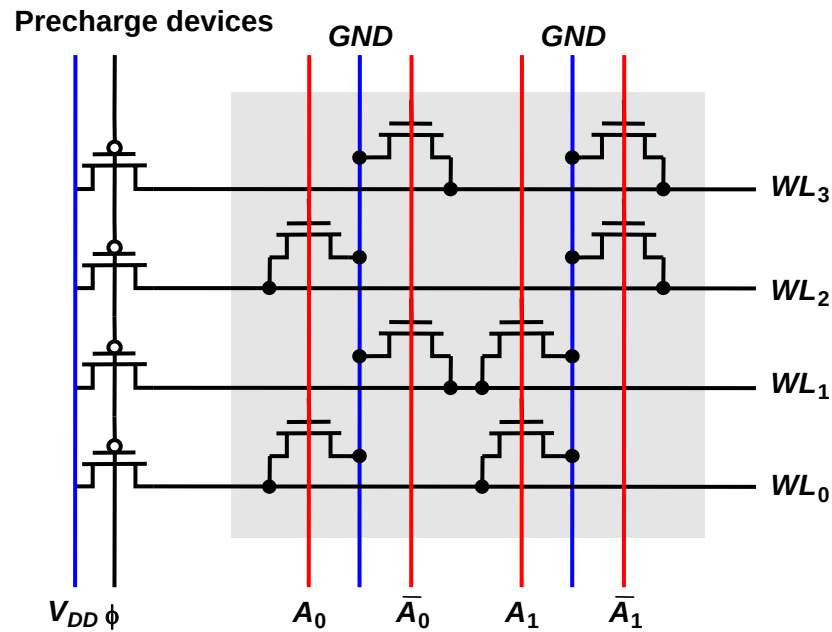
$$WL_{511} = \overline{A_0 + \bar{A}_1 + \bar{A}_2 + \bar{A}_3 + \bar{A}_4 + \bar{A}_5 + \bar{A}_6 + \bar{A}_7 + \bar{A}_8 + \bar{A}_9}$$

# Hierarchical Decoders

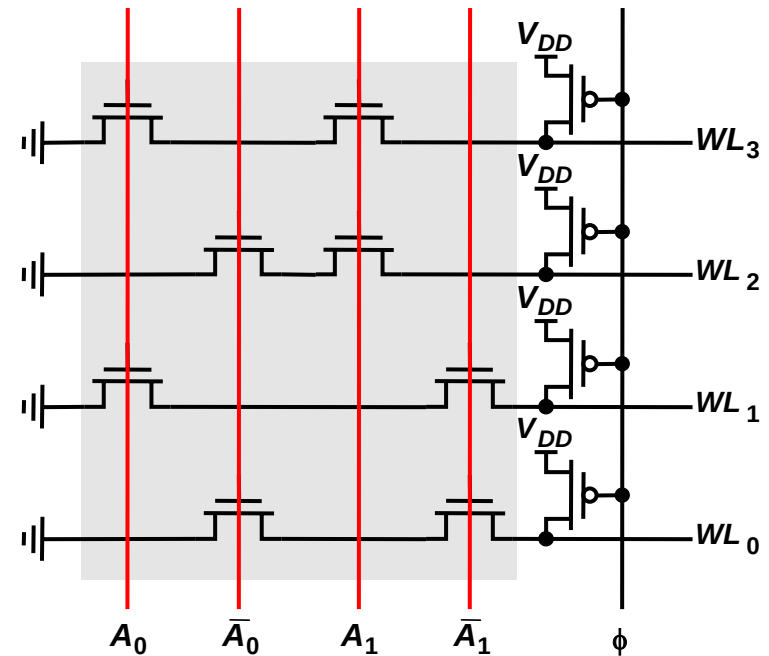
*Multi-stage implementation improves performance*



# Dynamic Decoders

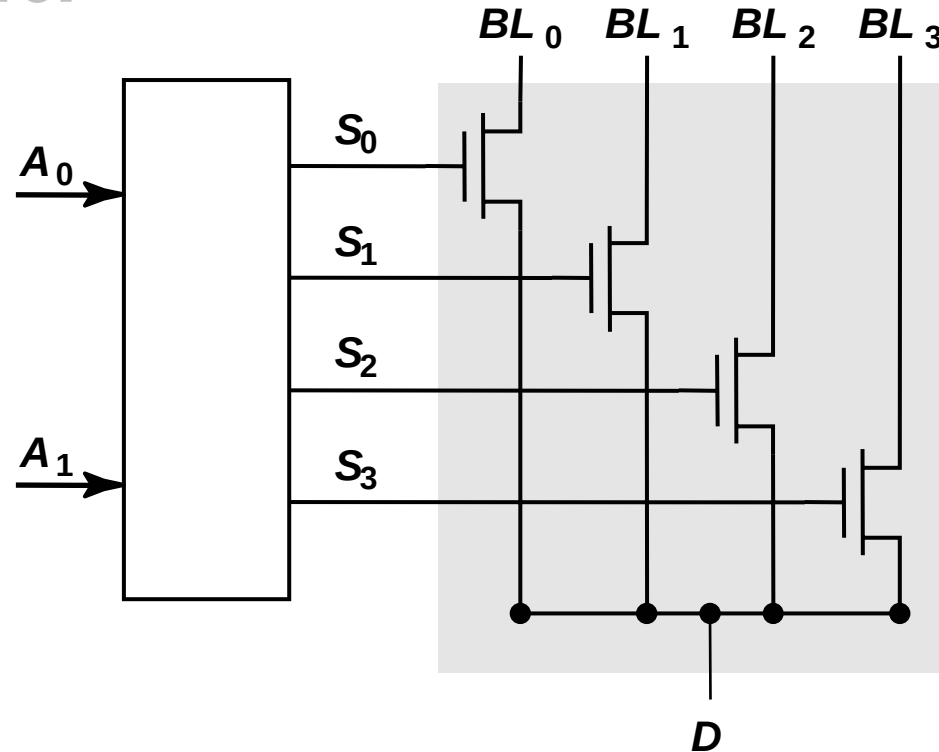


**2-input NOR decoder**



**2-input NAND decoder**

# 4-input pass-transistor based column decoder

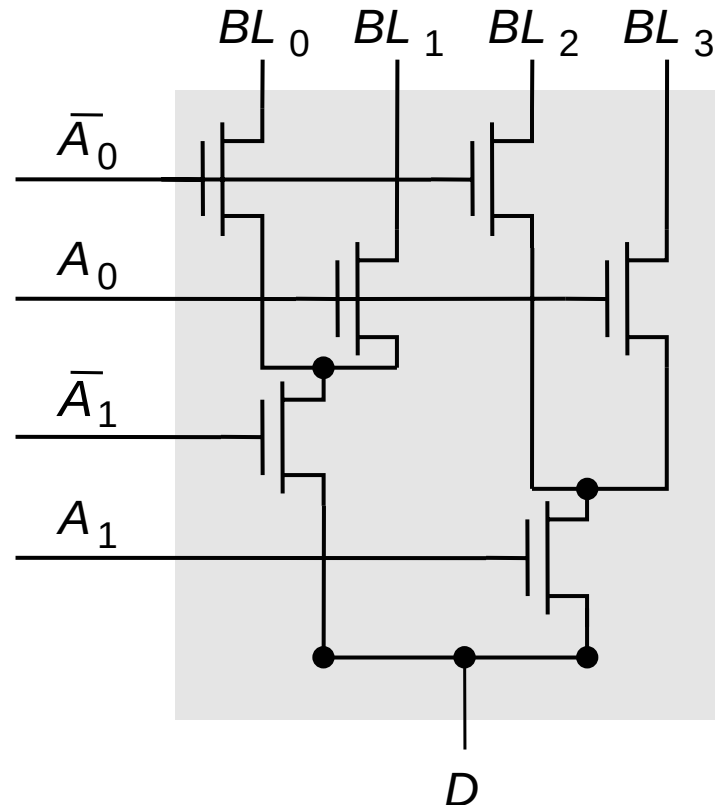


**Advantages:** speed ( $t_{pd}$  does not add to overall memory access time)

Only one extra transistor in signal path

**Disadvantage:** Large transistor count

# 4-to-1 tree based column decoder



Number of devices drastically reduced

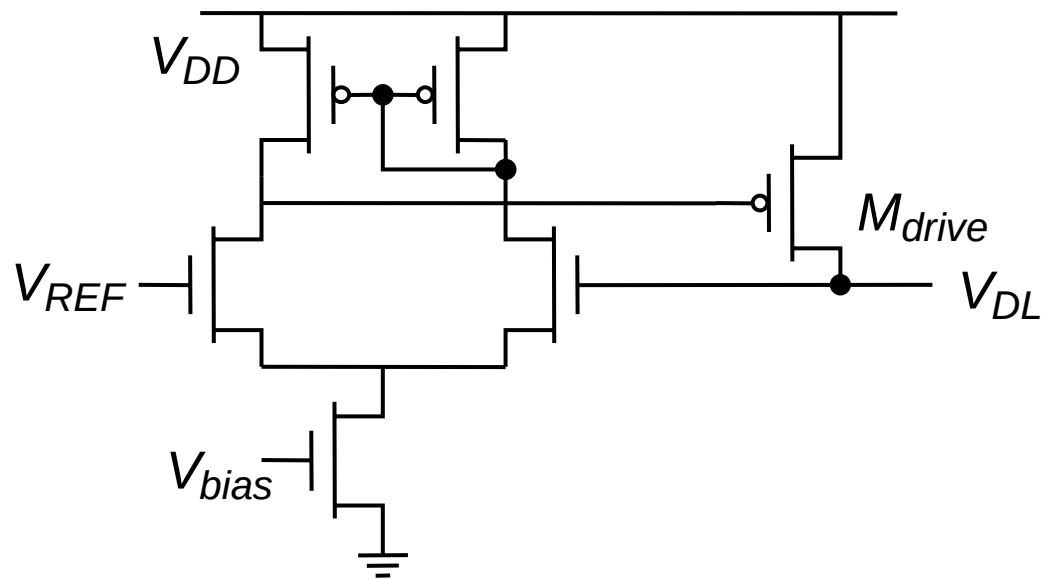
Delay increases quadratically with # of sections; prohibitive for large decoders

Solutions: buffers

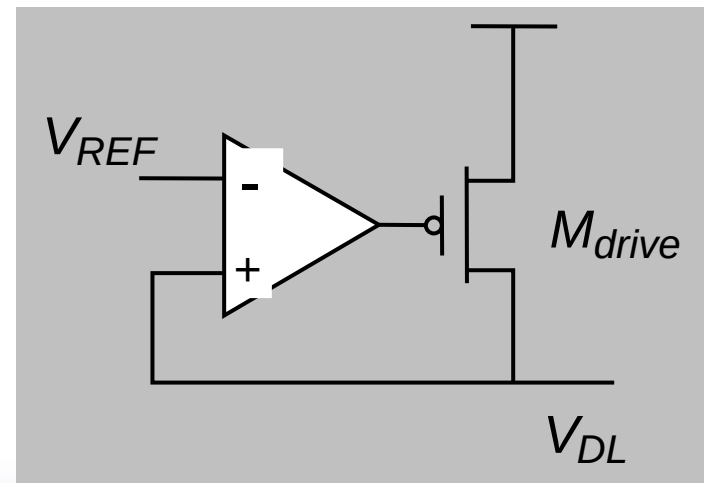
progressive sizing

combination of tree and pass transistor approaches

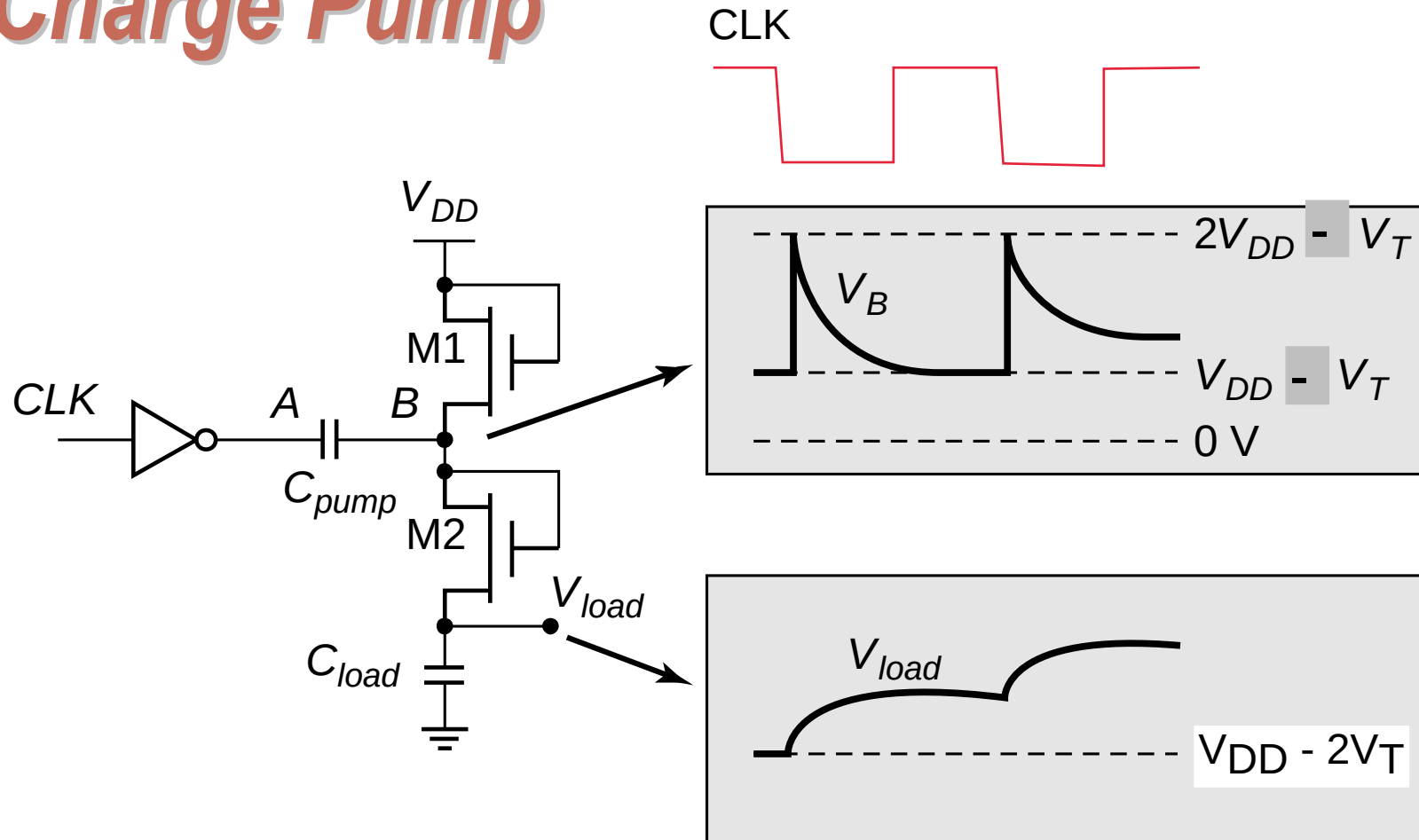
# Voltage Regulator



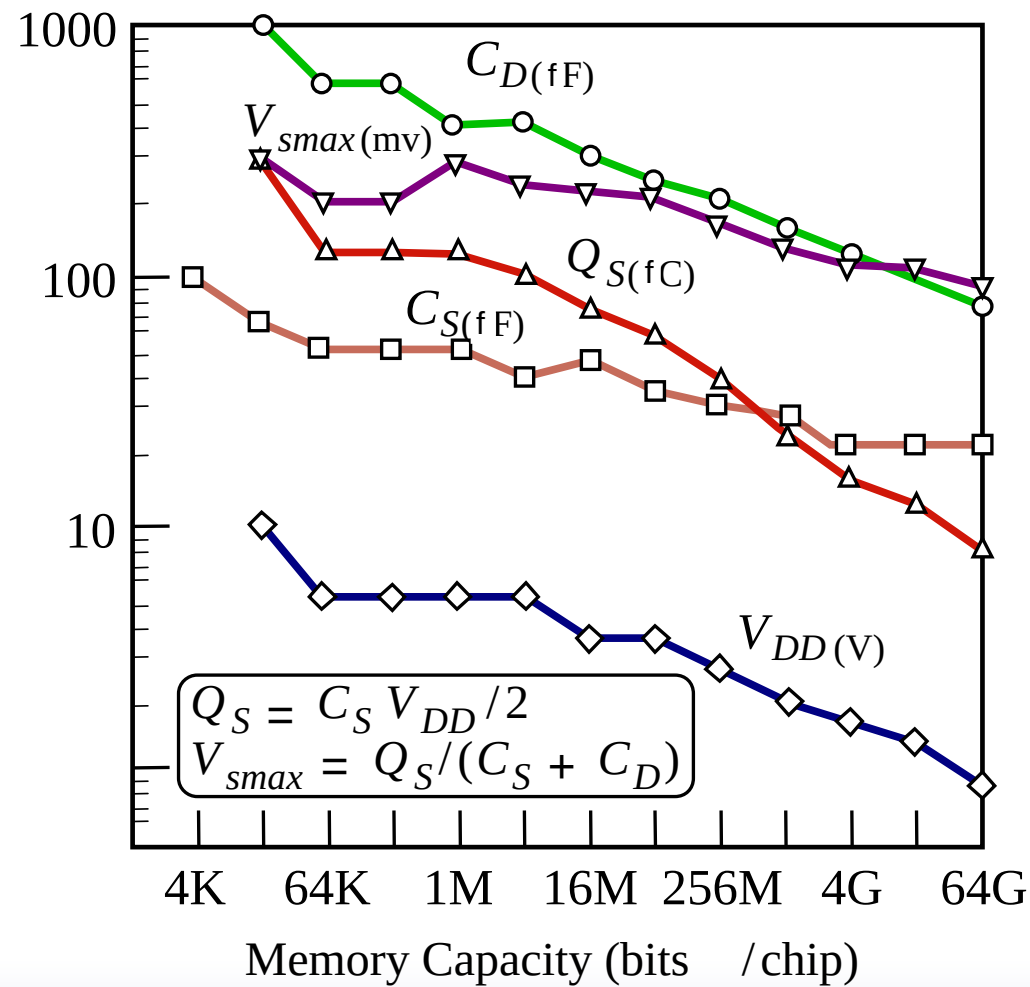
## Equivalent Model



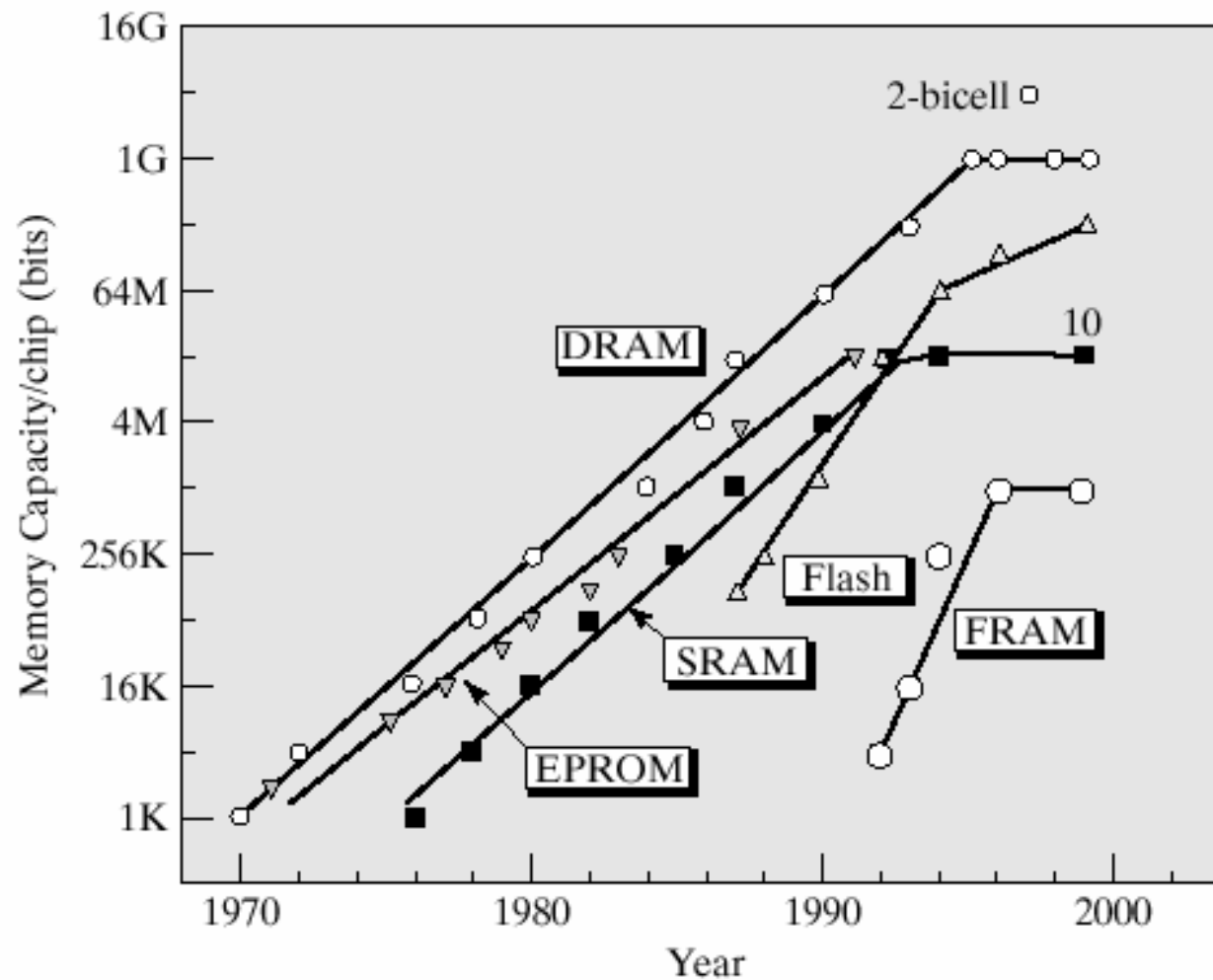
# Charge Pump



# Sensing-Parameters in DRAM



# Semiconductor Memory Trends (updated)



# *Trends in Memory Cell Area*

